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Module 9:

Halbleiterbauelemente

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Inhaltsverzeichnis

1 Introduction

Semiconductor devices form the backbone of modern electronics and are crucial to the functioning of numerous electronic devices such as computers, mobile phones and solar cells. These devices exploit the special properties of materials that are neither good conductors nor good insulators, but lie somewhere in between – so-called semiconductors. The following section discusses how they work, starting with the band model, which describes the energy states in solids. Based on various properties, other semiconductor materials and their applications are explained. A fundamental concept that plays a key role is the pn junction, which forms the basis for a wide range of components. In this context, the second part discusses various semiconductor components such as diodes and transistors and explains their functions.

Figure 1.1: **Sample photos of typical semiconductor components.** From left to right: field effect transistor, bipolar transistor, light-emitting diode, diode.

Learning objectives: Halbleiter

The students can

- explain the relationships between solids and the band model.
- describe processes within semiconductors.
- name different semiconductor materials and their properties.

1.1 Band model

The band model is a fundamental concept in solid state physics that describes the physical properties of solids. It provides a theoretical basis for understanding the electrical, optical and magnetic properties of materials. The model organises the energy states of electrons into so-called energy bands. These bands have a significant influence on the electronic behaviour of the material. The band model makes it possible to understand and explain complex phenomena such as conduction, insulation, semiconductor behaviour and the formation of surface and interface states. The following section explains the basic principles and takes a look at charge carrier transport.

To understand the band model, we must first establish a connection to Bohr's atomic model, which is familiar from school physics. In Bohr's atomic model, the energy levels of electrons in an atom are assumed to be discrete. According to this model, electrons are located on defined orbits around the atomic nucleus, which are referred to as shells. Each shell has a characteristic energy level. These energy levels are quantised in relation to the distance from the atomic nucleus, with electrons in the inner shells having lower energy levels than electrons in the outer shells. The relationship between the Bohr atomic model and the band model is shown in Figure 1.2.

Figure 1.2: **Connection between Bohr's atomic model and the band model.** Illustration of the relationship between the distance of electrons from the nucleus and the height of the corresponding discrete energy level. Left: Bohr's atomic model of a silicon atom. Right: Band model of a silicon atom with the possible energy levels.

In a single atom, the possible energy states are clearly defined. If at least two atoms are brought together so that they interact electrically, their energy states overlap. However, due to the Pauli principle, also known as the exclusion principle, two electrons cannot assume exactly the same state, which is why the states shift minimally in relation to each other. As the number of atoms increases, so does the number of different energy states in a solid. Since these levels are very close to each other, they are grouped together into energy bands. This relationship is shown in Figure 1.3.

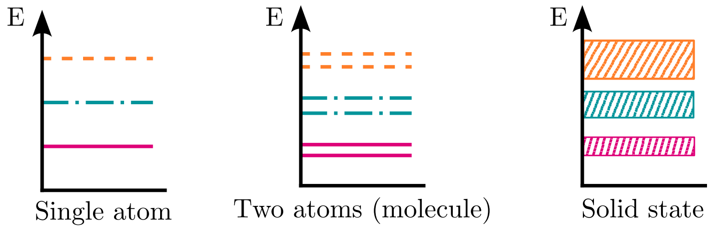


Figure 1.3: **Transition from energy levels to bands.** ZRelationship between the possible energy states depending on the number of atoms. From left to right: Energy states of electrons in a single atom, two atoms (molecule) and a solid.

Electrons can only assume fixed values within the energy bands. No charge carriers can move freely in the gaps between the bands. These gaps are referred to as the 'forbidden region' or band gap. The size of these gaps determines the electrical properties of the material. The energy difference between the bands corresponds to the energy absorbed or emitted during the absorption or emission of photons.

1.1.1 Classification of materials

Two bands are particularly relevant for charge transport and thus for electrical properties: the valence band (VB) and the conduction band (CB). The valence band is the band that contains the highest energy levels occupied by electrons when the material is at a temperature of 0 K. These electrons are tightly bound to the atomic nuclei and do not contribute to the electric current. The conduction band lies above the valence band and contains empty states in which electrons can be easily excited, for example by increasing the temperature. Electrons in the conduction band can move relatively freely through the material, thus enabling the flow of electric current. The three basic classes of conductors, semiconductors and insulators can be defined by the distance (band gap) between these two relevant bands. The following figure shows the band model of a material at 0 K, with the relevant quantities and designations.

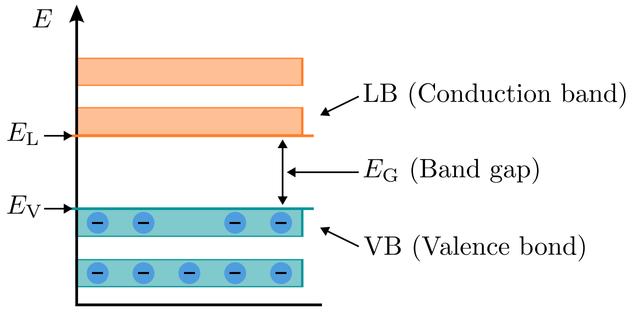


Figure 1.4: **Band model of a material at 0 K.** Representation of the energy bands with corresponding labelling and their alternative designations.

The occupation of the bands with electrons depends on other factors besides temperature. Conductivity can be influenced primarily by targeted contamination with foreign atoms, a process known as doping. The foreign atoms introduce free charge carriers and lead to additional energy levels within the band gap, from which, for example, light charge carriers can change bands (see section 1.3). In addition to the energy levels shown so far, the Fermi level is another important quantity. This indicates the energy value at which electrons have a probability of $1/2$ of being present. In an undoped semiconductor, the level lies midway between the valence and conduction bands, while in doped semiconductors, the level shifts towards the respective doping band.

Conductors are materials whose valence and conduction bands are adjacent or overlap, meaning that electrons can easily move between the bands. This proximity enables high conductivity, as electrons can move freely through the material. Metals such as copper and aluminium are typical examples of conductors.

Semiconductors have a small band gap that lies between that of conductors and insulators. This band gap is so large that no electric current flows in the material at low temperatures, as electrons do not have enough energy to be excited into the conduction band. However, conductivity can be significantly increased by raising the temperature or introducing foreign atoms. Semiconductors such as silicon and germanium are frequently used materials in the electronics industry.

Insulators have a large band gap, which means that the valence band is completely filled and the conduction band is empty. This makes it difficult for electrons to be excited into the conduction band, even with high energy input. Insulators such as glass and ceramics therefore exhibit very low conductivity.

The following are examples of band models with the valence and conduction bands of the three classes mentioned above.

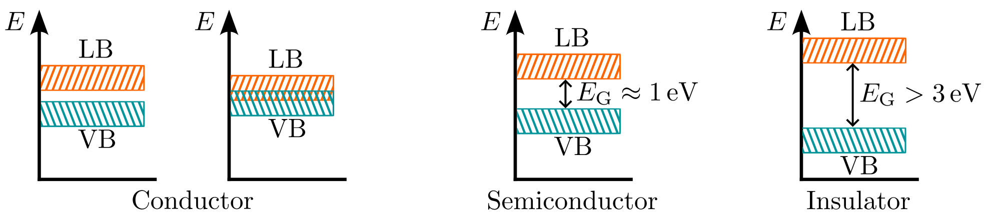


Figure 1.5: **Band model of various semiconductor materials.** From left to right: conductor without and with overlap, semiconductor and insulator.

Key point:

- Charge carriers can only occupy defined energy levels in the solid state.
- At $T = 0\,\mathrm{K}$, the valence band is the highest occupied energy level; the conduction band above it contains no free charge carriers.
- Materials can be categorised as band gap, conductors, semiconductors, and insulators.

1.2 semiconductor materials

The lattice structures of semiconductor materials play a decisive role in terms of their electronic and mechanical properties. The lattice structures of silicon and gallium arsenide (GaAs) are considered below. Silicon is the most widely used semiconductor material due to its high availability and the associated low price. In addition, it is very easy to process and its electrical properties can be easily influenced. Gallium arsenide, on the other hand, serves as an example of a compound semiconductor consisting of an element from the III and V main groups of the periodic table; accordingly, such a semiconductor is also referred to as a III/V compound semiconductor. The individual elements do not exhibit semiconductor behaviour; only when arranged in a specific way and with a specific atomic ratio does the semiconductor behaviour develop. Silicon has a diamond lattice structure in which each silicon atom is surrounded by four neighbouring atoms in a tetrahedral arrangement. This structure results in a stable and robust crystal structure with high mechanical stability. In addition, the arrangement allows high mobility of the free electrons in all spatial directions.

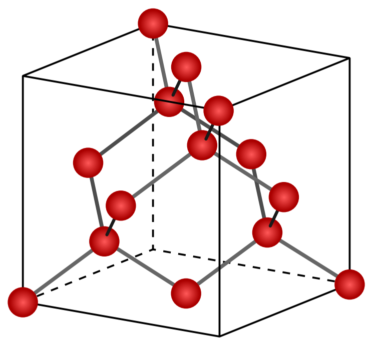


Figure 1.6: **Diamond lattice structure of silicon.** Resulting lattice structure of silicon.

In contrast, in gallium arsenide, the gallium and arsenic atoms are arranged in such a way that they form two interlocking cubic face-centred lattices, known as the zinc blende lattice structure. The resulting arrangement is identical to the diamond lattice structure, but with regular alternation between two types of ions. This structure allows for a small band gap, resulting in low excitation energy required to shift electrons between the valence and conduction bands.

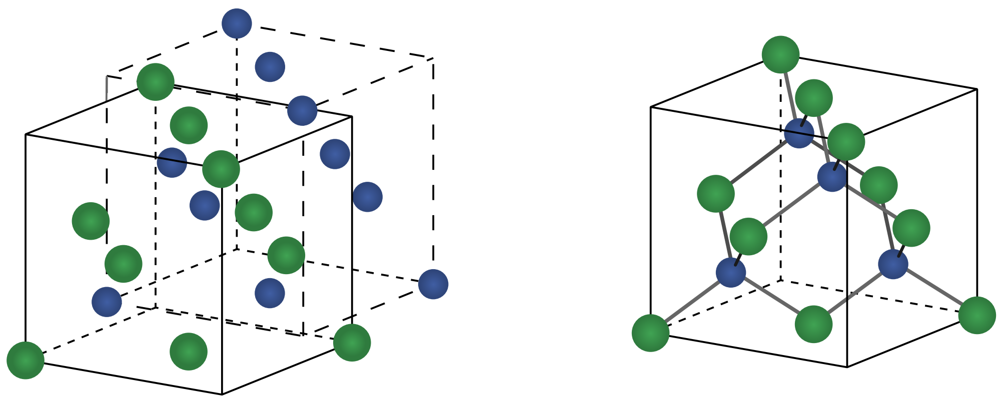


Figure 1.7: **Zinc blende lattice structure of GaAs.** Left: Two interlocking cubic face-centred lattices made of different materials. Right: Resulting zinc blende lattice structure.

Key point:

- Silicon is the most widely used semiconductor material.
- Compound semiconductors consist of two materials that exhibit semiconductor behaviour in certain combinations.
- Silicon is from the IV main group of the periodic table, compound semiconductors are typically from the III and V main groups.

The following table provides an overview of some of the most important semiconductor materials and compound semiconductors, as well as their key properties and applications. These materials play a crucial role in modern electrical engineering and are used in a wide range of applications, from microchips and solar cells to high-frequency circuits and LED lighting. The table contains information about the band gap, charge carrier density (intrinsic conductivity), electron and hole mobility, and characteristic properties and applications of each material. Mobility is the speed at which charge carriers move through a medium due to an electric field. Holes are quasi-partial states that are considered positive charge carriers. They are not actual positive charge carriers, but rather local areas that can be considered positively charged due to a missing electron.

Material	E_G (eV)	n $\left(\frac{1}{\mathrm{cm}^3}\right)$	μ_e $\left(\frac{\mathrm{cm}^2}{\mathrm{V \cdot s}}\right)$	μ_p $\left(\frac{\mathrm{cm}^2}{\mathrm{V \cdot s}}\right)$	Properties	Applications
silicon (Si)	1,1	$1 \cdot 10^{10}$	1500	450	Most common semiconductor materials	microchips, Solar cells, Sensors
Germanium (Ge)	0,7	$2 \cdot 10^{13}$	3900	1900	Previously frequently used in electronic devices	Transistors, infrared detectors
Gallium-arsenide (GaAs)	1,43	$2 \cdot 10^{6}$	8500	400	Direct bandpass, use at high frequencies	high-frequency circuits, LEDs, Laser diodes
Indium-phosphide (InP)	1,35	$1 \cdot 10^{16}$	5000	200	Direct band transition, high light absorption at wavelengths in the range of $1.3 - 1.55\,\mu$m	optoelectronics, solar cells
Gallium-nitride (GaN)	3,4	$1 \cdot 10^{-10}$	380	–	High thermal and chemical stability, high electronic breakdown field strength	power electronics, LED lighting, displays
Silizium-karbide (SiC)	3,0	$1 \cdot 10^{-7}$	500	–	Extremely high thermal stability, high electronic breakdown field strength	power electronics, high-temperature application

Table 1.1: **Properties of various semiconductor materials.** List of physical quantities such as the band gap (E_G), intrinsic conductivity (n) at $T = 300\,$K, electron mobility (μ_e) and hole mobility (μ_p).

1.3 Load carrier transport

At a temperature of $0\,$K (absolute zero), semiconductors such as silicon have a completely filled valence band with the energy level E_V and an empty conduction band with the energy level E_L. This means that all electrons are bound in the valence band and there are no free charge carriers. The band model shows a distinct band gap between the valence band and the conduction band. When the temperature is raised above $0\,$K, the mobility of the electrons in the crystal lattice increases. Some electrons in the valence band can gain energy through thermal excitation and transition to the conduction band, creating free electrons and holes. These free charge carriers contribute to the conductivity of the semiconductor.

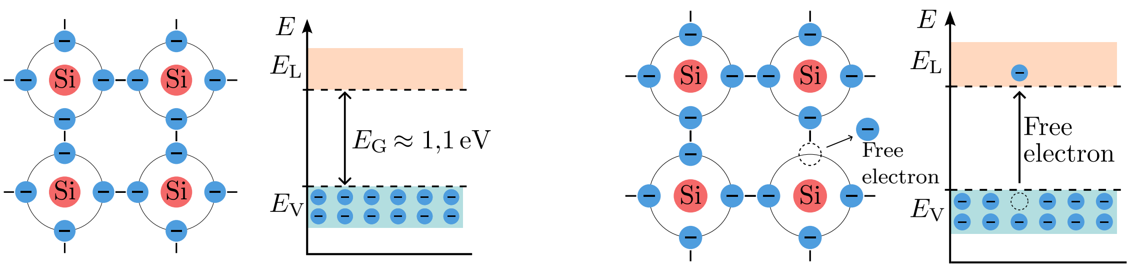


Figure 1.8: **Lattice structure and band model of Si.** Simplified 2D view of the silicon lattice. Left: Pure silicon at $T = 0\,$K. Right: Pure silicon at $T > 0\,$K.

When phosphorus is added to silicon, the phosphorus atom contributes five valence electrons (n-doped), one more than silicon. In this case, the excess electron is loosely bound to the crystal lattice and can be easily removed by external energy sources, making it a donor of free electrons. These electrons are located in the donor band (E_D), immediately below the conduction band. At temperatures above 0 K, thermal excitation gives the electrons enough energy to overcome the band gap and enter the conduction band. The electrons provided by the phosphorus facilitate this process. This increases the conductivity of the material.

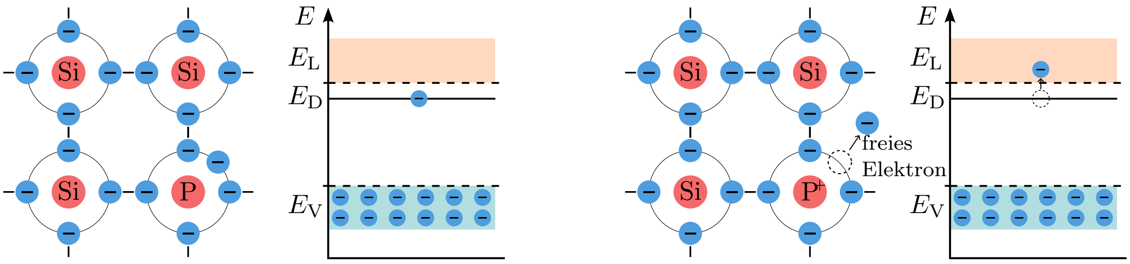


Figure 1.9: **Lattice structure and band model of n-doped Si.** Left: N-doped silicon at $T = 0\,\mathrm{K}$. Right: N-doped silicon at $T > 0\,\mathrm{K}$.

When boron is added to silicon (p-doped), the boron atom has only three valence electrons, one less than silicon. This creates a hole in the valence band of the crystal lattice, which acts as an electron acceptor. At 0 K, however, there are no thermally generated holes. At temperatures above 0 K, the thermal energy causes electrons to move from the valence band to the acceptor band (E_A), leaving holes in the valence band. These holes act as positive charge carriers and increase the conductivity of the material.

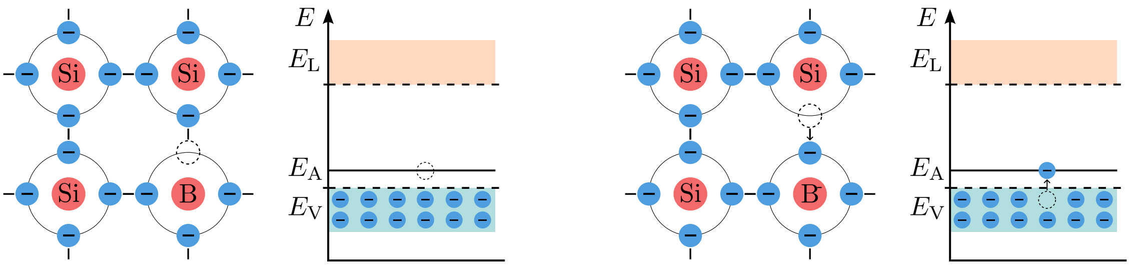


Figure 1.10: **Lattice structure and band model of p-doped Si.** Left: P-doped silicon at $T = 0\,\mathrm{K}$. Right: P-doped silicon at $T > 0\,\mathrm{K}$.

Key point:

- Thermal excitation can generate free electrons, which contribute to charge carrier transport.
- Doping is the targeted introduction of foreign atoms with more or less valence electrons than the starting material.
- Phosphorus can be used for n-doping and boron for p-doping.

In addition to the aforementioned processes within the crystal lattice, which enable charge carrier transport through additional charge carriers, two other important variables are drift current and diffusion current. Drift current in a semiconductor occurs due to the movement of charged particles under the influence of an external electric field. When an electric field is applied to a semiconductor, a force acts on the free charge carriers (electrons and holes), accelerating or decelerating them in the direction of the field, depending on their sign or charge. For electrons in the conduction band, this means that they drift towards the positive electrode (anode) under the influence of the electric field. For holes in the valence band, this means a drift towards the negative electrode (cathode). For better comparability, the hole is shown as a positive charge carrier in the following figure. The drift velocity of the charge carriers depends on the strength of the applied electric field and on the mobility of the charge carriers in the semiconductor material. It is important to note that the drift current only accounts for part of the total current in a semiconductor.

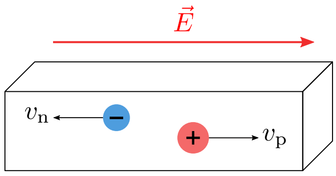


Figure 1.11: **Drift current within a semiconductor.** Representation with the velocity v of the charge carriers due to the electric field strength E.

The other part of the current is caused by diffusion, which is the movement of charge carriers due to concentration differences. In many semiconductor devices, such as diodes and transistors, drift and diffusion currents work together to determine the behaviour of the device. In diffusion current, free electrons or holes move from areas of high concentration to areas of low concentration, similar to the diffusion of particles in a concentration gradient. In the case of electrons in an n-doped semiconductor, electrons move from regions of high electron concentration to regions of low electron concentration. Conversely, in a p-doped semiconductor, holes move from regions of high hole concentration to regions of low hole concentration.

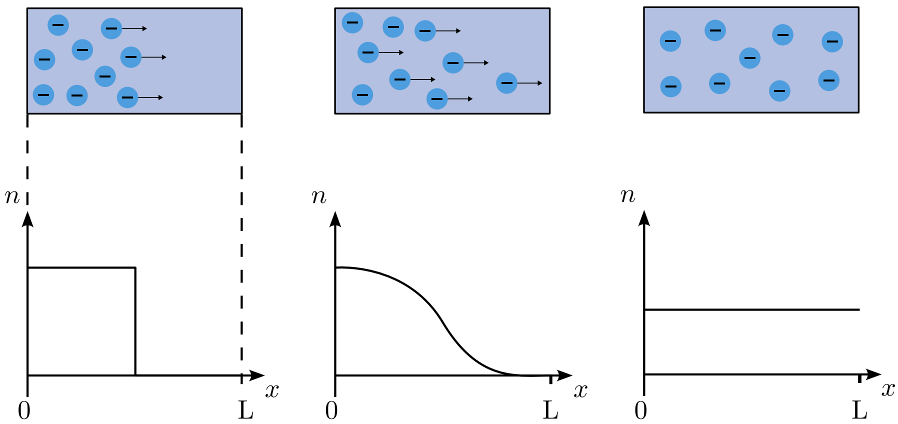


Figure 1.12: **Diffusion current within a semiconductor.** Diffusion of electrons in an n-doped semiconductor towards a lower concentration. With the length L of the semiconductor, the charge carrier concentration n and the position x.

Finally, this section examines charge transport in the band diagram when a voltage is applied. Until now, the band diagram has been analysed exclusively in a state of thermodynamic equilibrium. The general case is considered, in which a current flows through the semiconductor. A homogeneous, n-doped semiconductor serves as an example. Initially, no voltage is applied to the semiconductor, which means that no electric current flows. The band diagram over the location x therefore shows a curve similar to that in Figure 1.13 on the left, where possible edge effects at the contacts are not taken into account. The bands of a semiconductor without applied voltage are aligned horizontally. However, as soon as a voltage is applied, the band diagram shifts, causing the electrons to migrate towards lower energy.

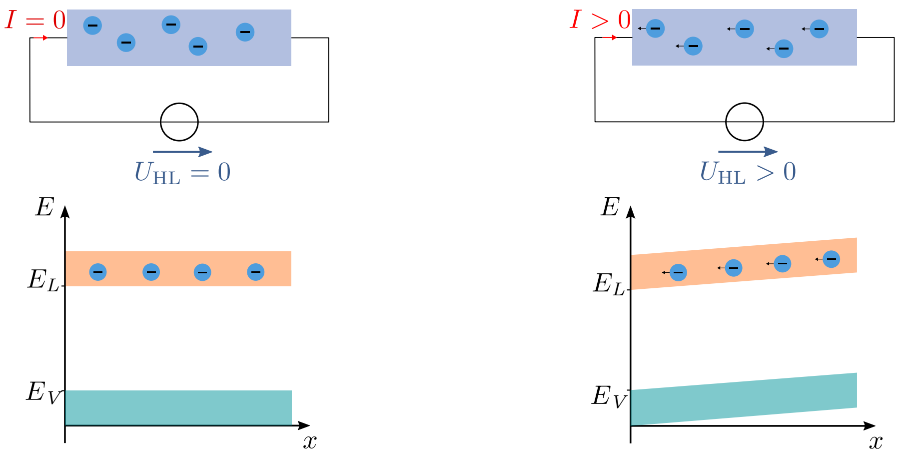


Figure 1.13: **Influence of voltages on semiconductors.** Left: Solid state and band model without applied stress. Right: Displacement of the band model due to applied stress.

Key point:

- Drift current is the transport of charge carriers due to an electric field.
- Diffusion current is the movement of charge carriers due to a difference in concentration.
- Drift current and diffusion current together make up the total current.
- An external stress causes a shift in the band model.

1.4 pn-junction

The pn junction is the central element of semiconductor devices such as diodes and transistors. It is created by connecting two differently doped semiconductor layers, a p-doped layer and an n-doped layer. The transition between these layers is called a pn junction. Free charge carriers diffuse in the pn junction: electrons from the n-doped layer diffuse to the p-doped layer and holes from the p-doped layer diffuse to the n-doped layer. This diffusion process leads to the formation of a so-called space charge region (SCR) in the junction region. This process is illustrated in Figure 1.14. The SCZ is a narrow region around the pn junction in which positive ions from the n layer and negative ions from the p layer remain after diffusion is complete. There are no free charge carriers in this zone, as the positive and negative charges neutralise each other. This creates an electric field ($\vec{E}$) that generates a drift current and suppresses the diffusion of further charge carriers. The space charge zone acts as a barrier layer and prevents current flow in the reverse direction.

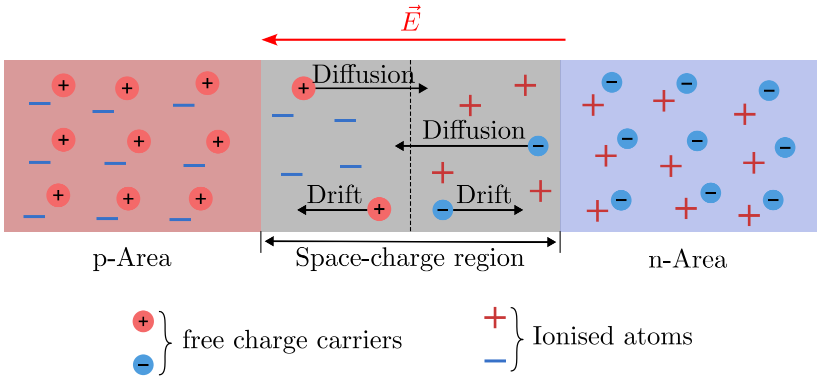


Figure 1.14: **Processes within the pn junction.** Drift and diffusion of charge carriers in the semiconductor, ions remaining in the RLZ.

When a voltage is applied in the forward direction (see Figure 1.15 on the left), the RLZ is reduced and the junction becomes conductive. Electrons from the n-doped side migrate to the p-doped side, while holes diffuse from the p-doped side to the n-doped side. This creates a current flow through the pn junction. Conversely, a voltage applied in the reverse direction increases the RLZ (see Figure 1.15 on the right). The free charge carriers are balanced by the voltage source. The pn junction is therefore a key element in semiconductor devices that controls the direction and strength of the current flow. Its properties are influenced by the doping concentration, the size of the space charge zone and the applied voltage.

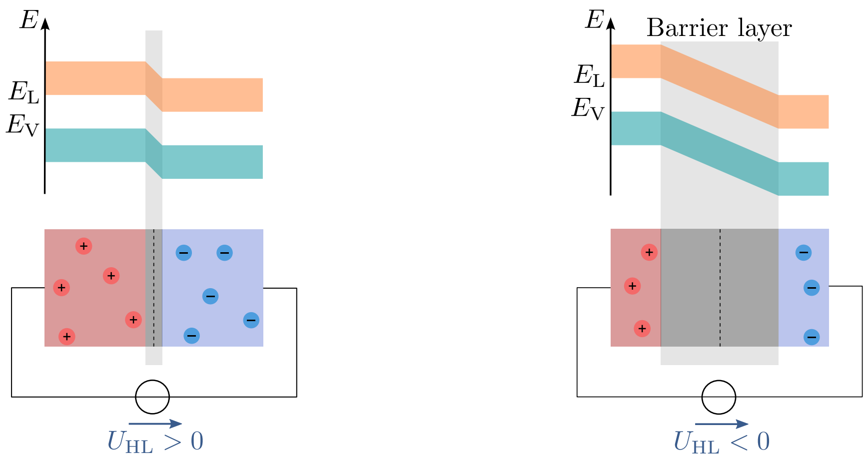


Figure 1.15: **Influence of voltages on pn junction.** Space charge zone at different applied voltages. Left: voltage in forward direction. Right: voltage in reverse direction.

Key point:

- The pn junction combines a p-doped and n-doped semiconductor layer.
- There are no free charge carriers in the transition region, known as the space charge zone.
- In the flow direction, the RLZ is reduced, and in the reverse direction, it is increased.

2 Electrical components

The following sections take a closer look at various types of semiconductor devices, including diodes, light-emitting diodes (LEDs), bipolar transistors and field-effect transistors. They cover the electrical behaviour, structure, key parameters and diverse applications of these devices.

Learning objectives: Halbleiterbauelemente

The students can

- Select components for various applications.
- Describe the structure of various semiconductor components.
- Determine operating points and calculate missing component values.
- Name applications for individual components.

2.1 Diode

The diode is the semiconductor component with the simplest structure, consisting of only one pn junction. This results in two terminals: the anode at the p-doped region and the cathode at the n-doped region. As already described in section 1.4, the pn junction allows current to flow in only one direction, which is why diodes are often used for rectification and voltage stabilisation.

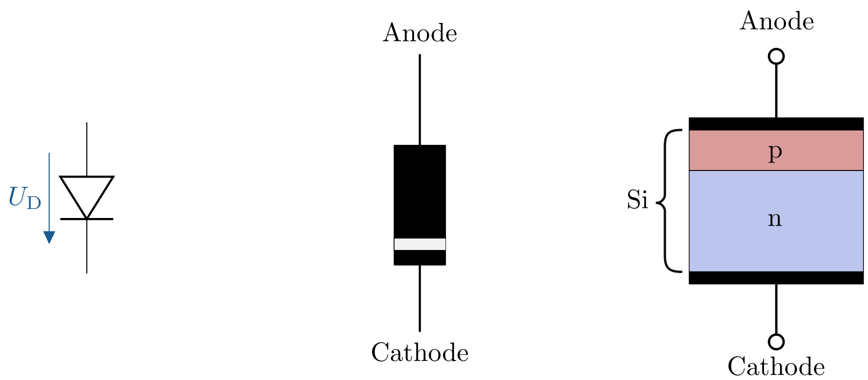


Figure 2.1: **Silicon diode.** From left to right: Circuit symbol, design and cross-section of a silicon diode.

2.1.1 Electrical behaviour

The electrical behaviour of diodes is essentially described by their characteristic curve, which is highly non-linear. This characteristic curve represents the current flowing through the diode as a function of the externally applied voltage. Characteristic areas of the characteristic curve are the forward, reverse and breakdown areas.

- **Forward bias range:** Operation in the forward bias range occurs when a positive voltage is applied in the direction of the diode, i.e. when the electrical potential at the anode is greater than that at the cathode (range to the right of the y-axis, see Figure 2.2). At a voltage above the barrier voltage (U_S), the diode current increases approximately exponentially. The barrier voltage is a material-specific voltage that is required at least to reduce the space charge zone sufficiently to allow current to flow. For silicon diodes, this is typically between 0.6 and 0.7 V.

- **Blocking range:** Occurs when the applied voltage is negative (range to the left of the y-axis). Hardly any current flows in this range because the resistance of the diode is very high.
- **Breakdown region:** At a voltage below the breakdown voltage (U_{BR}), a breakdown occurs and a very high current flows abruptly. If the diode is not specifically designed for breakdown, the high current leads to overload and consequently to destruction of the component.

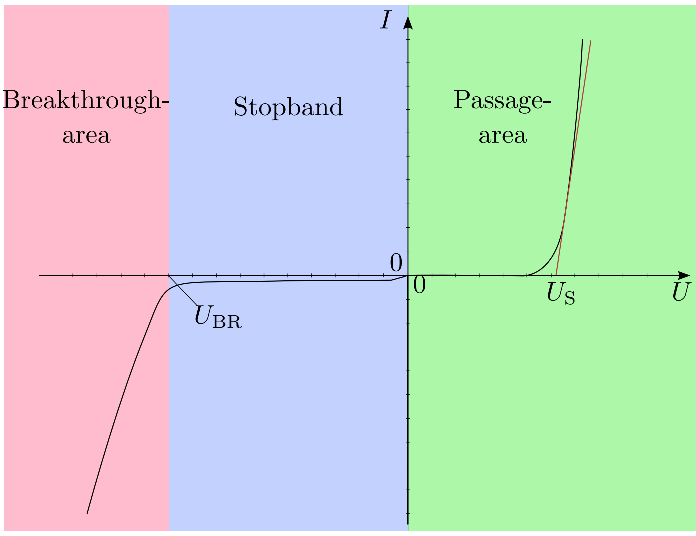


Figure 2.2: **Diode characteristic curve.** Relevant areas from left to right: breakthrough area, restricted area and passage area.

The equivalent circuit diagram of a silicon diode helps to better understand the behaviour of the diode in different operating states. It simplifies the real diode by modelling its essential characteristics.

Components of the equivalent circuit diagram:

- **Ideal diode (D):** Conducts in the forward direction without voltage loss and completely blocks in the reverse direction.
- **Lock voltage/threshold voltage (U_{S}/ U_{T0}):** This voltage is the value at which a measurable current flows through the real diode. In the equivalent circuit diagram, this is represented by an ideal voltage source.
- **Differential resistance (r_{D}):** The differential resistance (r_{D}) describes the resistance of the diode in the forward region. It is defined as the change in voltage (Δu_{D}) divided by the change in current (Δi_{D}) and models the non-linearity of the diode after reaching the gate voltage.

$$r_D = \frac{\Delta u_{\mathrm{D}}}{\Delta i_{\mathrm{D}}} \tag{2.1}$$

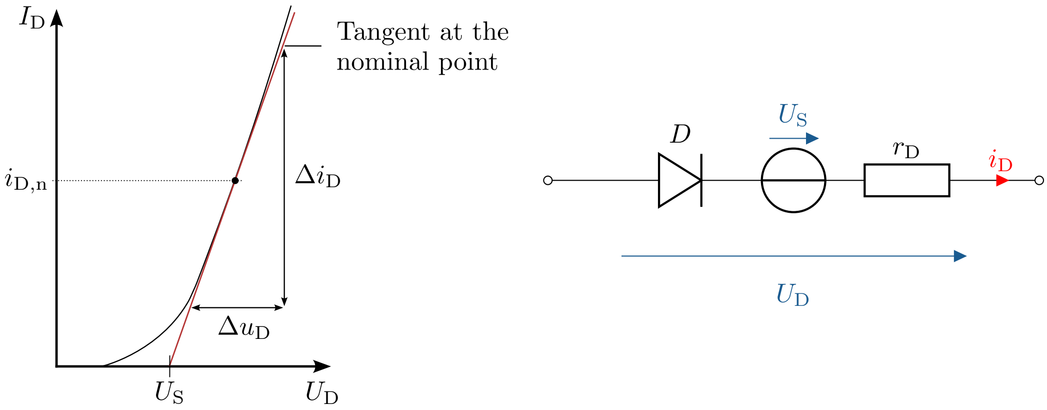


Figure 2.3: **Diode characteristic curve and equivalent circuit diagram.** Left: Diode characteristic curve with approximation for the differential resistance. Right: Diode equivalent circuit diagram with the ideal diode, voltage source and differential resistance.

Determining the operating point of a diode is an important step in circuit design, as it defines the stable operating state of the diode, taking into account the applied voltage and forward current. It is important to consider the maximum power dissipation of the diode to ensure that it does not overheat and become damaged. Both graphical and mathematical methods can be used to determine the operating point. The graphical method is based on the representation of the current-voltage characteristic curve of the diode and the load line of the circuit. The operating point is the intersection of these two lines. Both the operating point can be determined by the specified resistance, and the resistance can be determined based on the desired operating point. In both cases, the operating point should be below the asymptote for the maximum power dissipation of the diode.

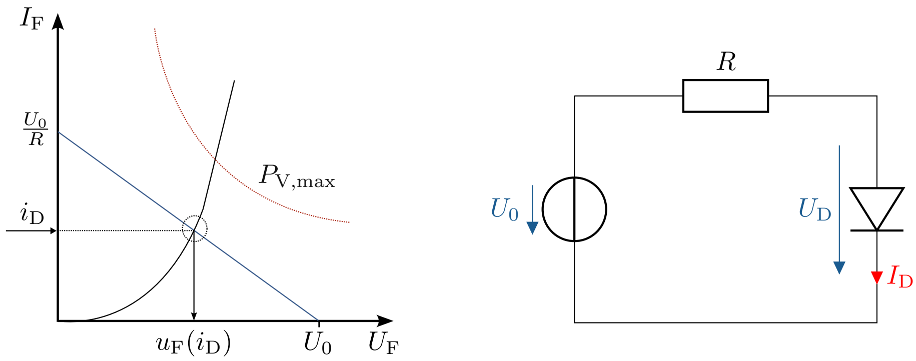


Figure 2.4: **Graphical determination of the operating point.** Left: Diode characteristic curve (black), resistance line (blue) and asymptote of maximum power dissipation (red) of the diode. Right: Circuit of the diode with series resistor.

The mathematical method is based on solving equations that describe the forward current of the diode and the voltage drop across the load. The operating point is determined by equating these two expressions. It is important to take into account the maximum power dissipation of the diode to ensure that it is operated within its specifications.

Key point:

- The characteristic curve of the diode is highly non-linear.
- The behaviour of the characteristic curve can be described in three areas: the breakdown, reverse and forward areas.
- Using an ideal diode, voltage source and resistor, the real diode can be approximated in different areas.

2.1.2 Structure

As already described in the introduction, the diode usually consists of a single pn junction. In addition to the simple silicon diode discussed so far, there are numerous other types of diodes. In this section, the structure discussed so far is compared with that of a Schottky diode. This is very common and does not require a pn junction. The pn diode consists of two semiconductor regions, namely the p-doped (positively charged) and the n-doped (negatively charged) region, which meet at a common interface. The p-side is called the anode and the n-side is called the cathode.

Typically, the pn junction consists of silicon or germanium. The electrons from the n-side recombine with the holes from the p-side at the interface, which leads to the formation of a space charge zone. This space charge zone forms the barrier for current flow in the reverse direction. A Schottky diode consists of a metal-semiconductor junction instead of a pn junction. During manufacture, no different dopants need to be introduced. Applying a suitable metal layer is sufficient. The semiconductor material is typically n-doped. The junction between the metal and the semiconductor forms a Schottky barrier that blocks the flow of current. With a suitable combination of materials, a space charge region can form at the interface, similar to a silicon diode.These diodes are optimised for fast switching operations and a low voltage drop in the forward direction.

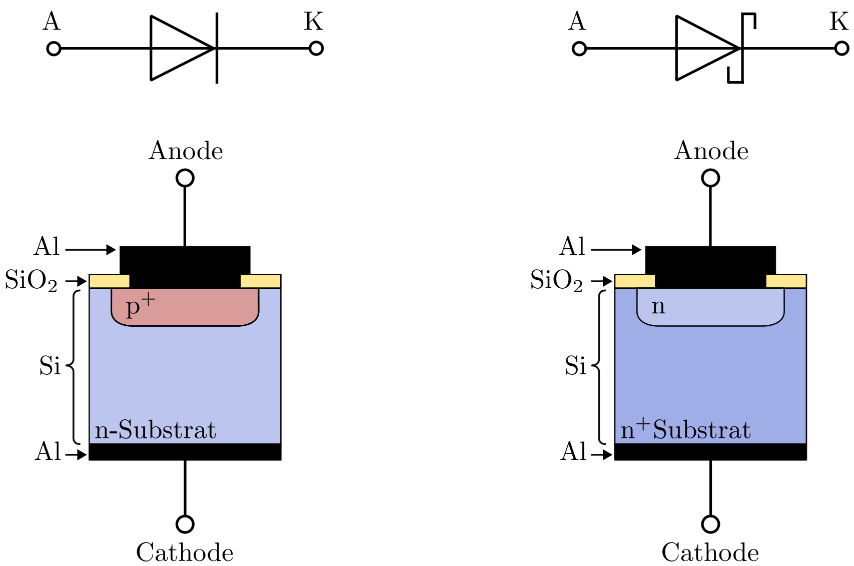


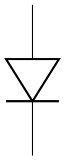
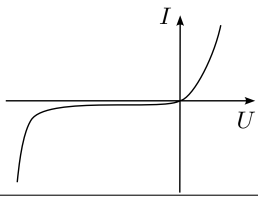
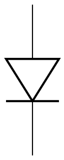
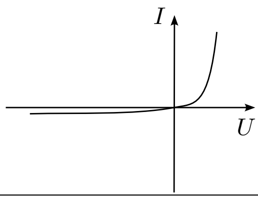
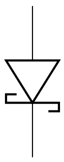
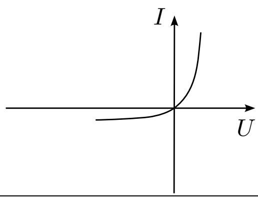
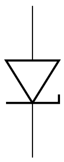
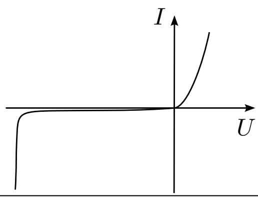
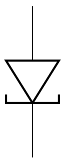
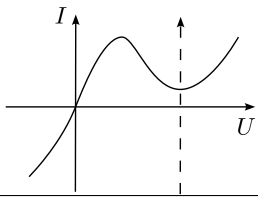
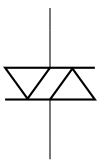
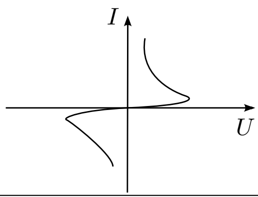
Abbildung 2.5: **Layer structure of diodes.** Structure and comparison of a classic silicon diode (left) and Schottky diode (right). Silicon dioxide (SiO_2) is a natural oxide of silicon and is used for insulation. Aluminium (Al) is a typical metal for electrical contacts.

Key point:

- Schottky diodes do not have a pn junction, but rather a Schottky contact.
- With a suitable combination of materials, an RLZ forms between the metal and the semiconductor.

2.1.3 Special diodes

In addition to the diodes mentioned above, there are a number of other variants that have different properties due to their design and enable corresponding applications. The following table provides an overview of typical diodes.

Designation	Symbol	Character. curve	Properties	Application
Rectifier diode			High forward current, high reverse voltage	rectification
Switching diode			low conduction resistance, high blocking resistance	short switching times
Schottkydiode			low forward voltage, low reverse voltage	HF rectifier, free-wheeling diode, switching power supplies
Z-Diode			defined breakdown voltage	stabilisation of voltages, limitation
Tunneldiode			negative differential resistance	de-attenuation of oscillating circuits, HF oscillator
Diac			Controlled breakdown	de-damping, trigger diode

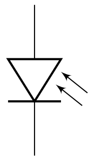
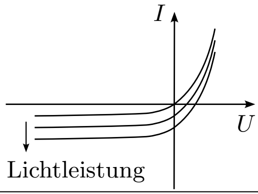
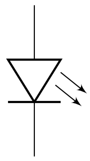
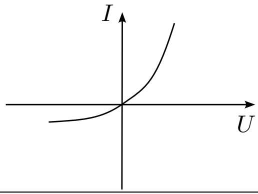
Designation	Symbol	Character. curve	Properties	Application
Photodiode			Current changes proportionally to light output	Photo receiver, measurement techno-logy, solar cells
LED, Laserdiode			Passage current generates optical radiation	lighting, radiation source

Table 2.1: **Overview of typical diodes.**

2.1.4 Applications/Basic circuits

This section introduces typical applications and basic circuits of diodes. Topics covered include single-phase rectifiers, bridge rectifiers, and series and parallel connections of diodes. Single-phase rectifiers and bridge rectifiers are essential circuits for converting alternating current into direct current.

Single-phase rectifier
A single-phase rectifier is a simple circuit for converting alternating current into direct current. This circuit typically consists of a single diode connected in series with the load. The main function of the single-phase rectifier is to allow only the positive half-waves of the alternating current to pass through, while blocking the negative half-waves.
During the positive half-wave of the alternating current, the potential at the anode is higher than that at the cathode, causing the diode to conduct in the forward direction. This allows current to flow through the diode and the connected load, resulting in a positive voltage across the load. During the negative half-wave of the alternating current, the potential at the anode is lower than that at the cathode, causing the diode to operate in the reverse direction and block the flow of current. As a result, the potential at the anode is lower than that at the cathode, causing the diode to operate in the reverse direction and block the flow of current. During the negative half-wave of the alternating current, the potential at the anode is lower than that at the cathode, so the diode operates in reverse direction and blocks the flow of current. As a result, no voltage is applied to the load. The following figure shows an example of the input voltage and the resulting voltage at the load. The difference between the two voltages (Δu) corresponds to the voltage drop across the diode. In the example, the voltage drop is approximated by the barrier voltage of a silicon diode.

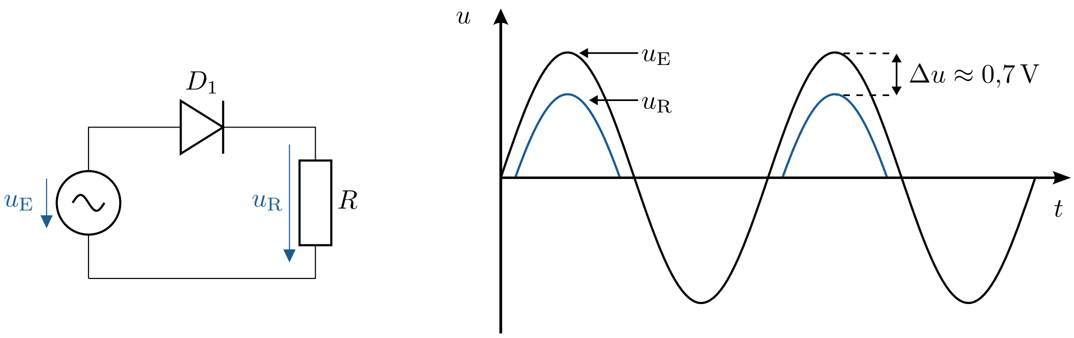


Figure 2.6: **Single-phase rectifier.** Left: Circuit diagram of the single-phase rectifier. Right: Voltage curve of the input voltage and load voltage of the single-phase rectifier.

The half-wave rectifier has the advantage of a simple circuit and low cost, making it suitable for basic applications where simple rectification is sufficient. However, this circuit also has significant disadvantages. Since only the positive half-waves of the alternating current are used, efficiency is low and the output voltage has a strong ripple, which is referred to as ripple voltage. This ripple can be reduced by additional filter and smoothing circuits. A capacitor may be sufficient to generate a more stable DC voltage.

Bridge rectifier
A bridge rectifier is a widely used circuit for converting alternating current into direct current. This circuit consists of four diodes arranged in a bridge configuration. The bridge rectifier utilises both half-waves of the alternating current, resulting in more efficient rectification than with half-wave rectifiers.

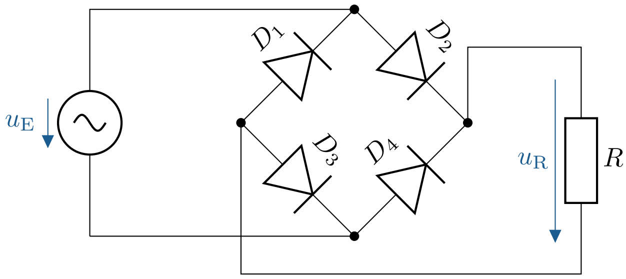


Figure 2.7: **Bridge rectifier circuit.** Circuit of a bridge rectifier with applied load R.

During each half-wave of the alternating current, two of the four diodes conduct and form a path for the current to flow through the load. In the positive half-wave, two diodes conduct the current in one direction (Figure 2.8 left), and in the negative half-wave, the other two diodes conduct the current in the same direction through the load (Figure 2.8 right).

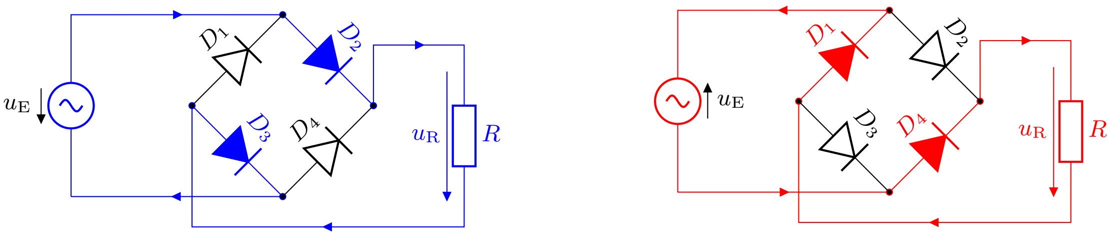


Figure 2.8: **Current paths in the bridge rectifier.** Left: Current path during the positive half-wave of the input voltage. Right: Current path during the negative half-wave of the input voltage.

The resulting voltages, as a consequence of the two current paths shown previously, are illustrated below. This behaviour of the bridge rectifier means that the voltage across the load always has the same polarity, which produces a rectified output voltage.

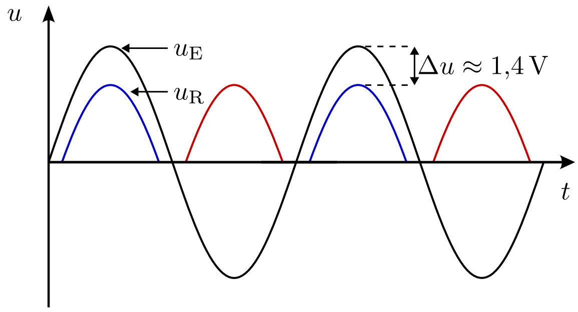


Figure 2.9: **Bridge rectifier signal waveforms.** Input voltage and load voltage curve of the bridge rectifier.

The output voltage has significantly shorter interruptions compared to a half-wave rectifier, resulting in a more stable and smoother DC voltage. However, the bridge rectifier requires more diodes than a half-wave rectifier, leading to higher costs and a greater voltage drop.

Key point:

- Diodes are used in rectifiers to convert alternating voltages into direct voltages.
- In half-wave rectifiers, only one polarity of the half-waves is passed through.
- Bridge rectifiers use both polarities of the input voltage.

Series and parallel connection
Series and parallel connection of diodes are fundamental methods for adjusting the electrical characteristics of circuits. These configurations are often used to meet the voltage and current requirements of diodes in various applications.

- **Series connection:** In a series connection, several diodes are connected in sequence. The maximum permissible total voltage is the sum of the reverse voltages of the individual diodes. This makes it possible to block higher voltages than with a single diode. This configuration is often used in high-voltage applications.
- **Parallel connection:** In a parallel connection, several diodes are connected in parallel, with the anodes and cathodes connected to each other. This configuration increases the maximum current carrying capacity, as the current is divided between the diodes. This method is often used to distribute the current and reduce the load on individual diodes.

By combining diodes in series or parallel circuits, the electrical properties of the overall circuit can be specifically adjusted to meet specific requirements.

2.2 Bipolar transistor

The bipolar junction transistor (BJT) is an essential component of many electronic circuits. It consists of three layers of semiconductor materials with alternating doping: the emitter (E), the base (B) and the collector (C). The functioning of a bipolar transistor is based on the control of the current flow between the collector and emitter by the base current. Depending on the order of the doping, a distinction is made between npn and pnp transistors. The following figure shows a simplified cross-section, the representation using diodes according to the pn junctions and the respective circuit symbol.

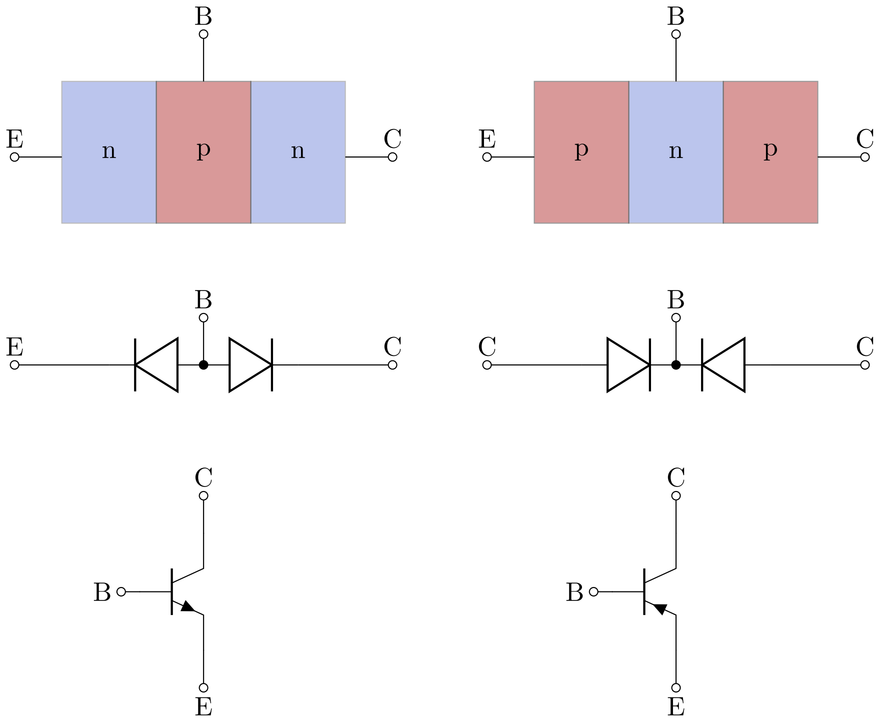


Figure 2.10: **Overview of bipolar transistors.** Simplified representation of the cross-section, logic and circuit symbols of npn and pnp transistors.

2.2.1 Electrical behaviour

The following section examines the electrical behaviour of the more commonly used NPN transistor based on the differently doped semiconductor layers and the associated band model. The behaviour can also be applied to the PNP transistor accordingly. Depending on the structure, two PN junctions are formed between the three layers. The blocking behaviour of the two diodes in the unconnected state can be seen both in the formed space charge zone (SCZ) and in the high energy difference between the bands in the different areas. The different heights of the composite band model can be explained by the Fermi levels of the individual areas. In the p region, the Fermi level is lower than in the n region. However, since the Fermi level is constant across the individual regions, this results in the steps shown in the band model.

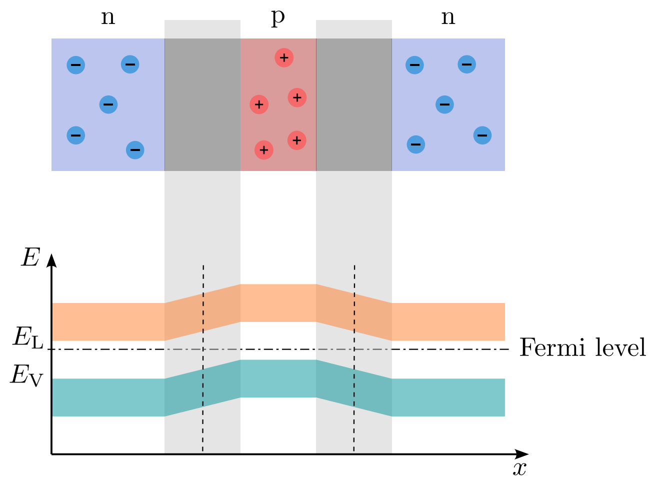


Figure 2.11: **Behaviour of an unconnected bipolar transistor.** Cross-section of an unconnected NPN transistor and associated band model.

If a positive voltage is applied between the collector and emitter, the pn junction between the base and emitter is in the forward direction, but the junction between the base and collector is in the reverse direction, which is why no current flow is possible. The charge carriers provided by the voltage source reduce the RLZ between the base and emitter. Conversely, the electrons flowing away from the collector side increase the associated RLZ. In addition, the following figure shows the strong shift in the bands due to the applied voltage.

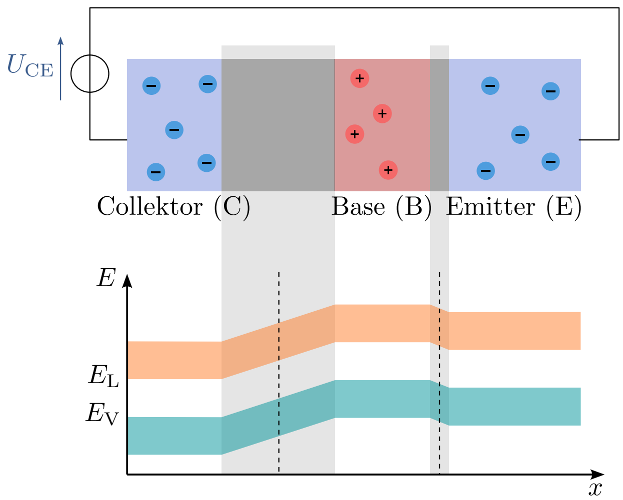


Figure 2.12: **Bipolar transistor with collector-emitter voltage.** Cross-section of an NPN transistor and associated band model with a positive voltage between the collector and emitter.

Applying an additional positive voltage between the base and emitter completely breaks down the RLZ in the direction of the emitter. The applied voltage corresponds to the barrier voltage of the pn junction. As a result, electrons can pass from the emitter to the base. These electrons are located immediately in front of the RLZ of the collector-base junction, which is in the forward direction. Consequently, the electrons are accelerated by the electric field formed and can completely traverse the semiconductor. The potential difference between the base and collector, which the electrons can easily traverse, can also be seen in the band model.

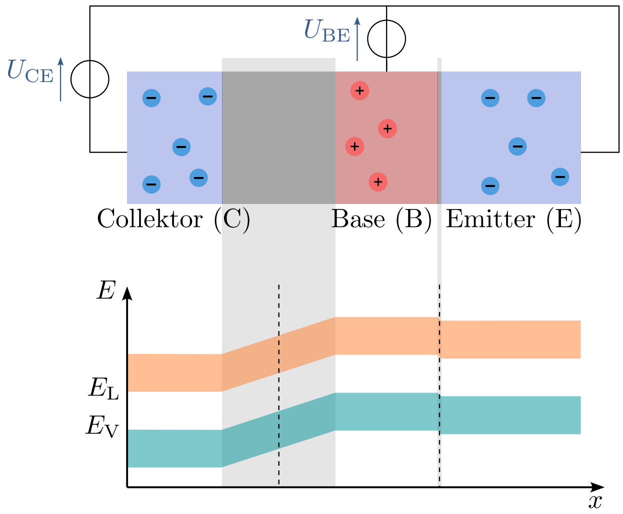


Figure 2.13: **Wired bipolar transistor.** Cross-section of an NPN transistor and associated band model with a positive voltage between the collector and emitter and between the base and emitter.

Merke:

- Bipolartransistoren weisen zwei pn-Übergänge auf.
- Es gibt drei Anschlüsse: den Kollektor (C), die Basis (B) und den Emitter (E).

The following figure shows the relevant currents and voltages for npn and pnp transistors. The voltages are usually referenced to the emitter potential and the currents are plotted in the direction of the transistors. In the case of the npn transistor, this results in a negative value for the emitter current, which represents the sum of the two partial currents.

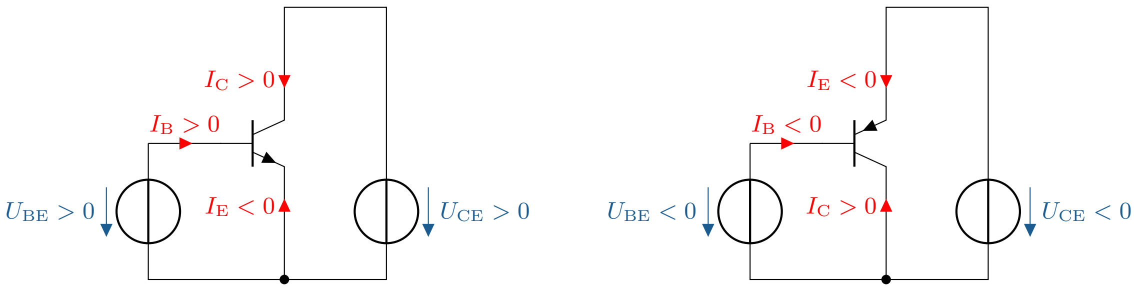


Figure 2.14: **Currents and voltages in bipolar transistors.** Relevant currents and voltages for npn and pnp transistors.

To describe the electrical behaviour, the collector and emitter currents as well as the two aforementioned voltages are considered. This results in four couplings: the currents as a consequence of the corresponding voltages and, in each case, between the two currents and the two voltages. First, the input characteristic curve, the base current I_B as a result of the base-emitter voltage U_{BE}, is considered. As can be seen in the following figur, the curve corresponds to that of a diode characteristic curve. The characteristic curve shown is representative of a defined collector-emitter voltage U_{CE}. If this voltage varies, the curve is stretched or compressed. This is also referred to as the input characteristic curve field. As already known from the diode characteristic curve, the differential resistance

r_{BE} can be determined from the slope at the operating point.

$$r_{\mathrm{BE}} = \frac{\Delta u_{\mathrm{BE}}}{\Delta i_{\mathrm{B}}} \tag{2.2}$$

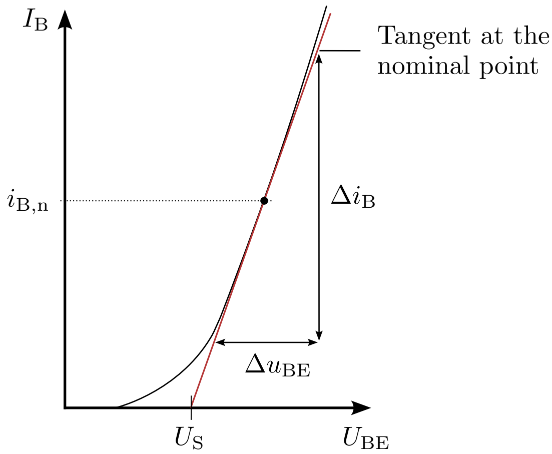


Figure 2.15: **Input characteristic curve of an NPN transistor.** Identification of relevant variables.

Key point:

- The input characteristic curve $I_{\mathrm{B}} = f(U_{\mathrm{BE}})$ is similar to that of a diode.
- A change in U_{CE} leads to a shift in the characteristic curve.

In the output characteristic curve, the collector current I_{C} is considered as a result of the collector-emitter voltage U_{CE}. In the active range, the influence of U_{CE} on I_{C} is negligible and a linear relationship can be assumed. In real components, the curve is flatter than shown in the illustration. Depending on the base current, the output characteristic field shown (see Figure 2.16) results. The differential resistance r_{CE} can be determined using the slope of the characteristic curve.

$$r_{\mathrm{CE}} = \frac{\Delta u_{\mathrm{CE}}}{\Delta i_{\mathrm{C}}} \tag{2.3}$$

Below the active region, at low values of U_{CE}, both diodes are switched in the forward direction and the transistor goes into saturation with a decreasing collector current. If the characteristic curves are extended into the negative region to the left of the y-axis, they intersect at a point on the x-axis. The underlying dependency is referred to as the Early effect and the voltage as the Early voltage (U_{Early}).

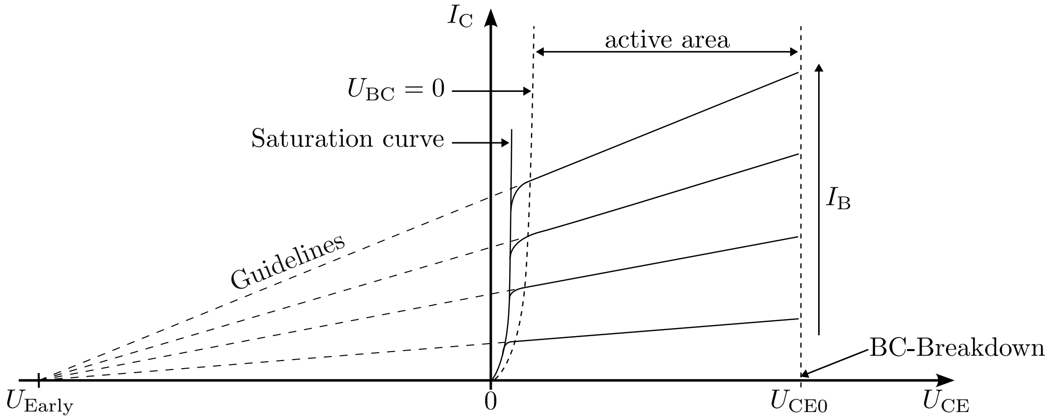


Figure 2.16: **Output characteristic curve of an npn transistor.** Marking of relevant areas and sizes as well as graphical determination of the early voltage.

Key point:

- The output characteristic curve represents $I_C = f(U_{CE})$.
- In the active range, the curve flattens out and the influence of U_{CE} on I_C decreases.
- I_B influences the height of the characteristic curve.

Another relationship considers the effect of the control current I_B on I_C. The representation is therefore referred to as the current control characteristic field. This results in the parameter direct current gain B and the differential current gain factor β at the operating point.

$$B = \frac{I_C}{I_B} \tag{2.4}$$

$$\beta = \frac{\Delta i_C}{\Delta i_B} \tag{2.5}$$

In the last case, the feedback characteristic field, the relationship between U_{BE} and U_{CE} is considered. The couplings depend on the set base current. Analogous to the description for current, the differential DC voltage gain D at the operating point can be considered for voltages.

$$D = \frac{\Delta u_{BE}}{\Delta u_{CE}} \tag{2.6}$$

Combining the four characteristic curves results in the following four-quadrant characteristic curve (see Figure 2.17). The representation of all couplings makes it possible, for example, to graphically transfer a desired operating point at the output to the input and to determine the parameters required for this.

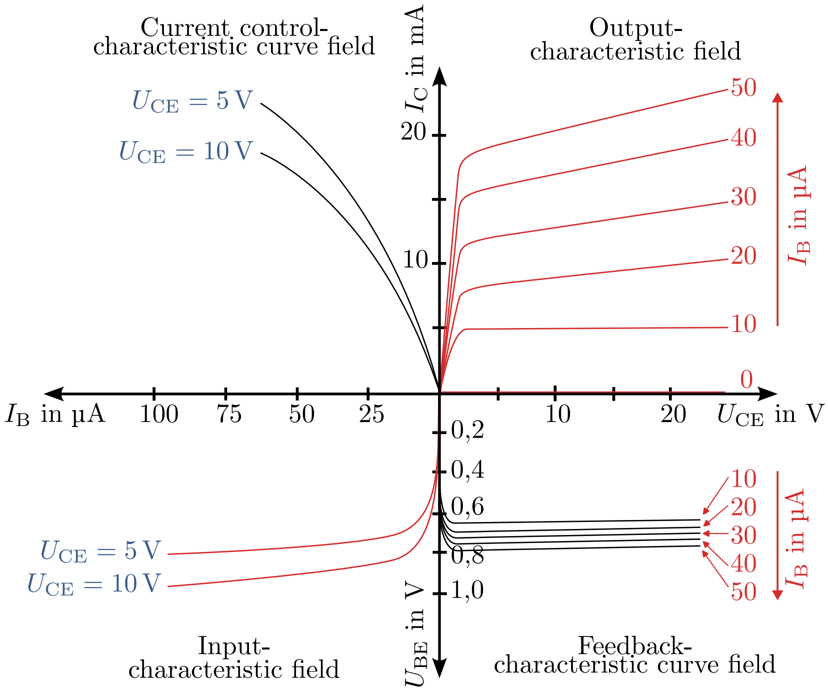


Figure 2.17: **Four-quadrant characteristic curve of an npn transistor.** Presentation of all relevant areas and sizes.

The differential variables used for mathematical description at the operating point are also called small-signal parameters. Using the small-signal parameters, the following equations can be set up and represented in an equivalent circuit diagram. This can be used for small-signal operation up to approx. 1 MHz.

$$u_{BE} = r_{BE} \cdot i_B + D \cdot u_{CE} \tag{2.7}$$

$$i_C = \beta \cdot i_B + \frac{u_{CE}}{r_{CE}} \tag{2.8}$$

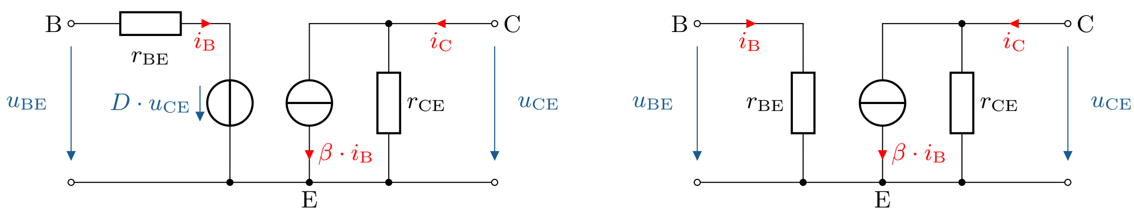


Abbildung 2.18: **Equivalent circuit diagram of a bipolar transistor.** Left: complete small-signal equivalent circuit diagram with all relevant components and variables. Right: simplification by neglecting the differential current gain.

Since D is typically very small, the voltage source can usually be neglected.

Key point:

- The four-quadrant characteristic curve field shows the current-voltage characteristic curves at the input and output as well as the couplings between them.
- The equivalent circuit diagram can be drawn up using the differential variables.

2.2.2 Operating point and small-signal behaviour

The characteristic curves shown above can be used to determine the operating point and the necessary parameters. To do this, let us consider the circuit shown in Figure 2.19.

Example 2.1

The supply voltage U_V is 20 V. Half the supply voltage should be applied across the resistor R_C, which represents an ohmic load, and a current of 10 mA should flow.

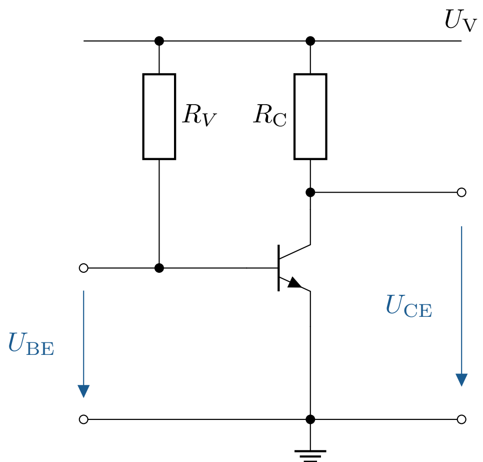


Figure 2.19: **Example of a transistor circuit for determining relevant parameters.** This circuit can be used to determine the relevant parameters.

In the first step, the resistance line can be plotted in the output characteristic field, starting from the operating voltage. The second necessary point represents the desired operating point A. Starting from this point, the operating point can be transferred to the other quadrants.

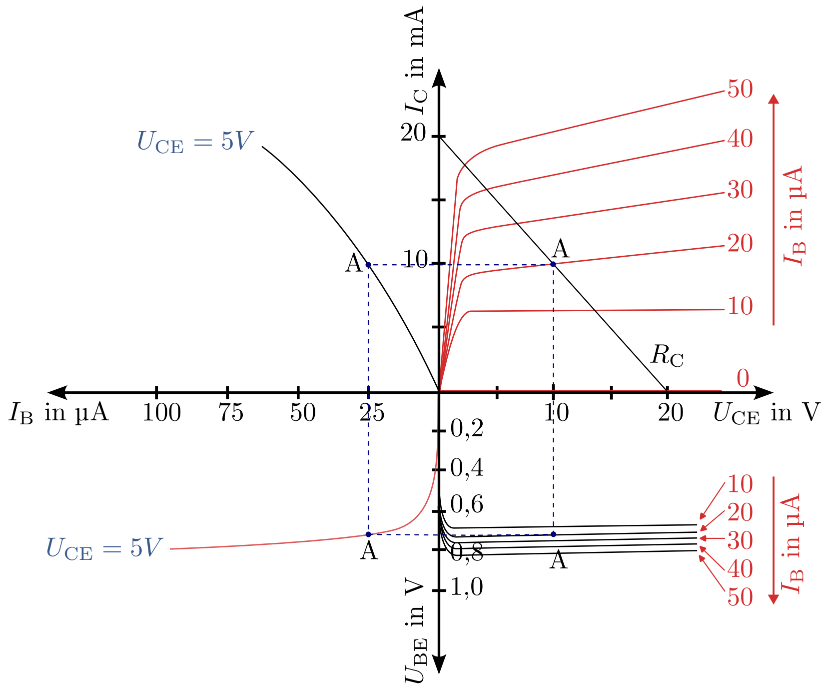


Figure 2.20: **Four-quadrant characteristic curve with operating point A.** Presentation of all relevant areas and dimensions, including operating points.

The following operating points and parameters result in the respective quadrants.

Output (at: $I_B = 20\,\mu\text{A}$):

$$U_{CE} = 10\,\text{V}$$
$$I_C = 10\,\text{mA}$$
$$R_C = \frac{U_{RC}}{I_{RC}} = \frac{U_V/2}{I_C} = \frac{10\,\text{V}}{10\,\text{mA}} = 1000\,\Omega$$

Current control:

$$I_C = 10\,\text{mA}$$
$$I_B = 25\,\mu\text{A}$$
$$B = \frac{I_C}{I_B} = \frac{10\,\text{mA}}{25\,\mu\text{A}} = 400$$

Entrance:

$$I_B = 25\,\mu\text{A}$$
$$U_{BE} = 0{,}72\,\text{V}$$
$$R_V = \frac{U_{RV}}{I_{RV}} = \frac{U_V - U_{BE}}{I_B} = \frac{20\,\text{V} - 0{,}72\,\text{V}}{25\,\mu\text{A}} = 771{,}2\,\text{k}\Omega$$

Feedback:

$$U_{BE} = 0{,}72\,\text{V}$$
$$U_{CE} = 10\,\text{V}$$
$$D = \frac{U_{BE}}{U_{CE}} = \frac{0{,}72\,\text{V}}{10\,\text{V}} = 0{,}072$$

In the event of fluctuations in the base current, whether due to an analogue input signal or interference, the resulting operating points or ranges can also be represented using the four-quadrant characteristic curve field. The behaviour can be seen in the following figure. An increase in the base-emitter voltage (U_{BE}) also causes an increase in the base current (I_B). In accordance with the coupling via the current control characteristic curve, the collector current (I_C) also increases; only the collector-emitter voltage (U_{CE}) decreases in this example.

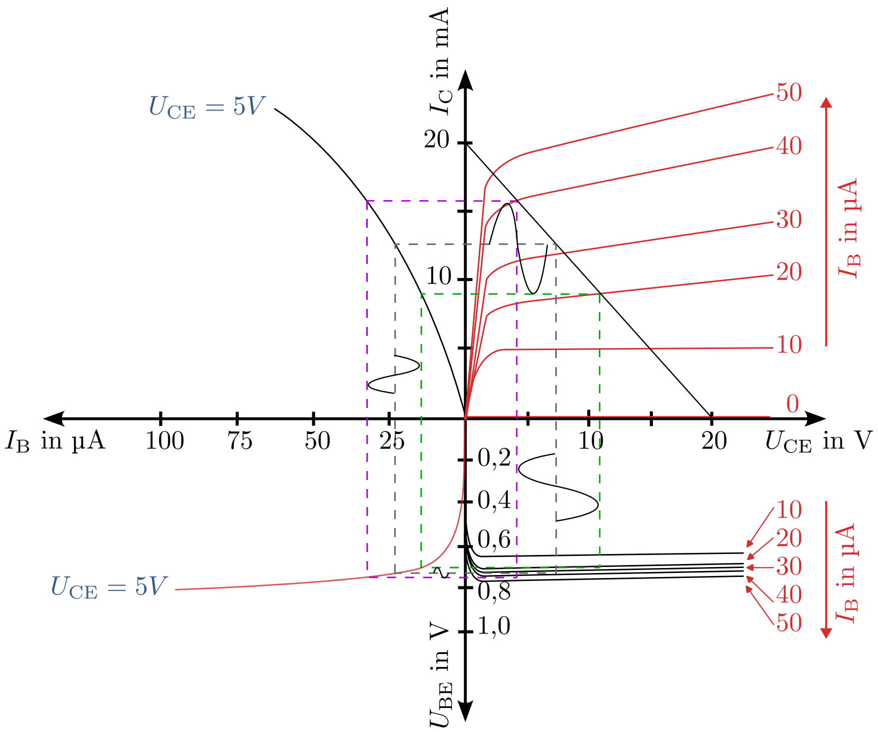


Figure 2.21: **Four-quadrant characteristic curve field with AC input signal.** Presentation of all relevant areas and sizes.

2.2.3 Structure

Until now, the bipolar transistor has been depicted as stacked according to the left side of Figure 2.22, with the differently doped regions lying directly on top of each other and the base in the middle. This simplified representation cannot be directly implemented in reality, as the different dopings are introduced one after the other into a surface. Accordingly, the lower layers cannot be directly electrically contacted as shown on the left. A typical implementation can be seen on the right in Figure 2.22. In an n-doped semiconductor, a p-doped area is subsequently introduced for the base. Within this p-well, further dopant atoms are introduced, creating a smaller n-well for the emitter. The desired layer structure with n-p-n can thus be seen below the emitter region. The different regions can be electrically connected by means of conductive connections on the surface.

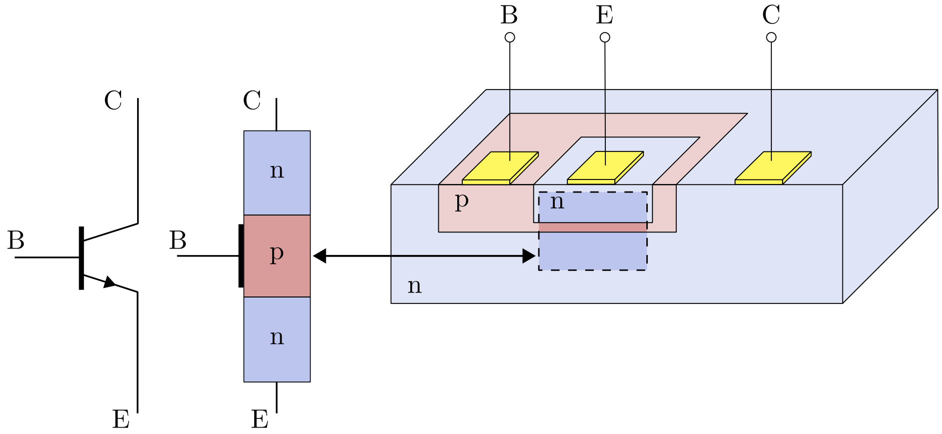


Figure 2.22: **Structure of an NPN bipolar transistor.** Left: Ideal schematic structure of an NPN bipolar transistor, based on its circuit diagram. Right: Actual structure of a planar NPN transistor.

2.2.4 Application/Basic circuits

Basic circuits
With bipolar transistors, a distinction is made between three basic circuits: emitter, collector and base circuits. The naming is based on the reference potential of the input and output voltage.

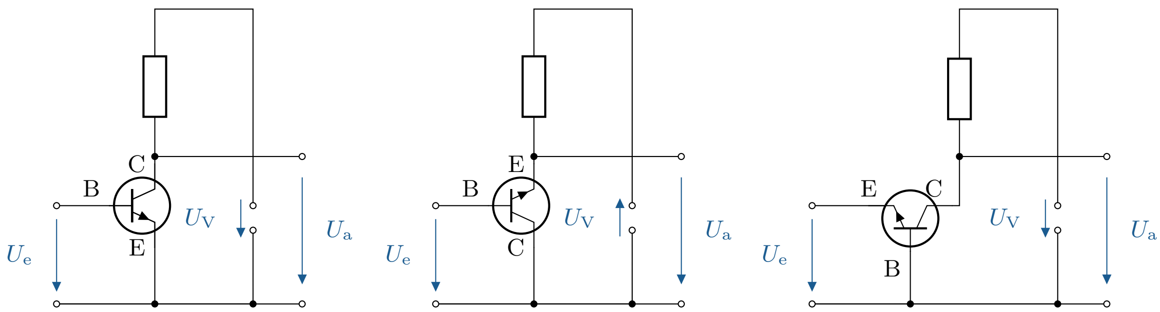


Figure 2.23: **Basic circuits npn bipolar transistor.** From left to right: emitter, collector and base circuit.

The most common use is the emitter circuit shown above, in which the transistor is used as an inverting amplifier. The inverting behaviour refers to a phase shift of -180° of the output signal relative to the input signal. The following figure shows the corresponding equivalent circuits; the resulting properties are summarised in Table 2.2. The differential resistance r_{CE} is parallel to the current source $\beta \cdot i_B$. Since it is very large, it is not taken into account in the simplified consideration.

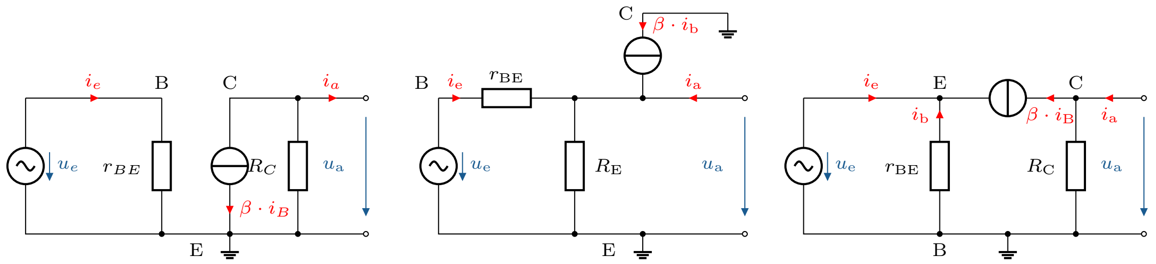


Figure 2.24: **Replacement circuit diagrams for the basic circuits.** From left to right: Emitter, collector and base circuit of npn transistors.

Properties	Emitter circuit	Collector circuit	Base circuit
Input resistance r_e	medium $1\,\mathrm{k\Omega}$	high $> 100\,\mathrm{k\Omega}$	low $50\,\Omega$
Output resistance r_a	medium $10\,\mathrm{k\Omega}$	low $50\,\Omega$	high $100\,\mathrm{k\Omega}$
Current amplification v_i	high 100	high 100	low < 1
Voltage amplification v_u	high 100	low < 1	high 100
Power amplification v_p	very high$1\,\mathrm{k}$	high 100	high 100
Phase rotation φ_u	antiphase 180°	in phase 0°	in phase 0°

Table 2.2: **Electrical properties of basic circuits.** Typical values of the electrical properties of various basic circuits of npn transistors.

Transistor as a switch

If small electrical loads need to be switched quickly and without contact, a bipolar transistor can be used. This assumes two different states: conductive and non-conductive in the collector-emitter path. In the conducting state, the C-E path has low resistance and represents a closed switch; in the blocking state, the opposite is true. The state can be controlled via the base-emitter path, resulting in an emitter circuit. The following figure shows the corresponding circuit and the voltage curves.

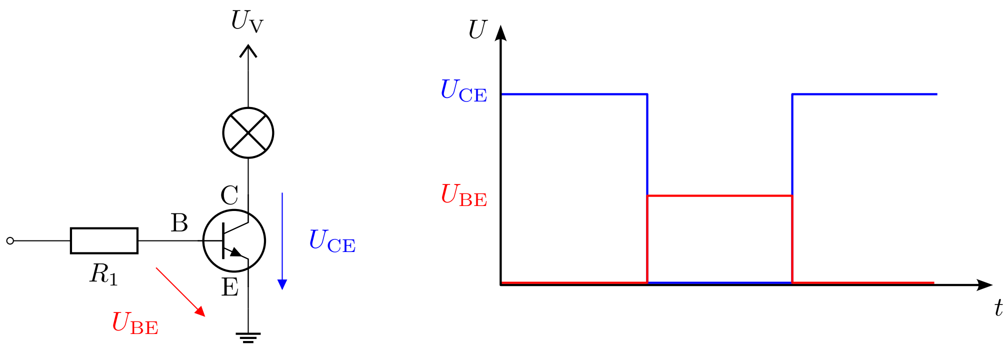


Figure 2.25: **Transistor as a switch.** Left: Circuit diagram with lamp as load in the collector-emitter path. Right: Voltage curves with U_CE as a result of U_BE.

Key point:

Compared to an electromechanical switch in the form of a relay, transistors have significantly smaller installation space and lower price. Due to the contactless design, there is also no contact bounce, which leads to a longer service life.

Darlington transistor

If the necessary current amplification of a single transistor is too low, it is possible to use a Darlington transistor or a Darlington circuit. This is based on two transistors, whereby the emitter of the first transistor feeds the base of the second transistor. This is based on two transistors, with the emitter of the first transistor feeding the base of the second transistor. Approximately, the resulting current amplification is the product of the individual amplifications:

$$B \approx B_1 \cdot B_2 \tag{2.9}$$

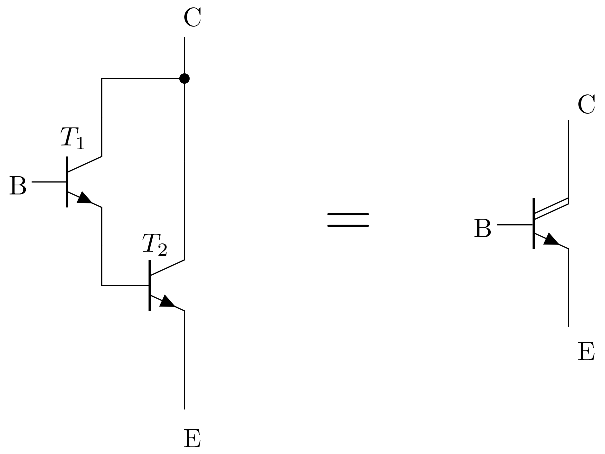


Figure 2.26: **Darlington transistor.** Circuit and circuit symbol of an NPN Darlington transistor.

2.3 Field-effect transistor

In addition to the bipolar transistor, there is another widely used type of transistor known as the field-effect transistor (FET). Its operating principle differs fundamentally from that of the bipolar transistor. While in a bipolar transistor a pn junction is set to the conductive state by a control terminal, in a field-effect transistor an electric field influences the distribution of free charge carriers in the semiconductor. This changes the resistance in the component. In FETs, the three terminals are called gate (G), source (S) and drain (D). The gate, also known as the control electrode, which influences the electric field, can be designed in different way, resulting in different types of transistors. The following figure shows an overview of the different types of transistors, including bipolar transistors. FETs are mainly divided into three types: junction field-effect transistors (JFETs), self-conducting MOSFETs (enrichment type) and self-blocking MOSFETs (depletion type). The suffix MOS stands for metal-oxide-semiconductor, the original layer structure of the control electrode.

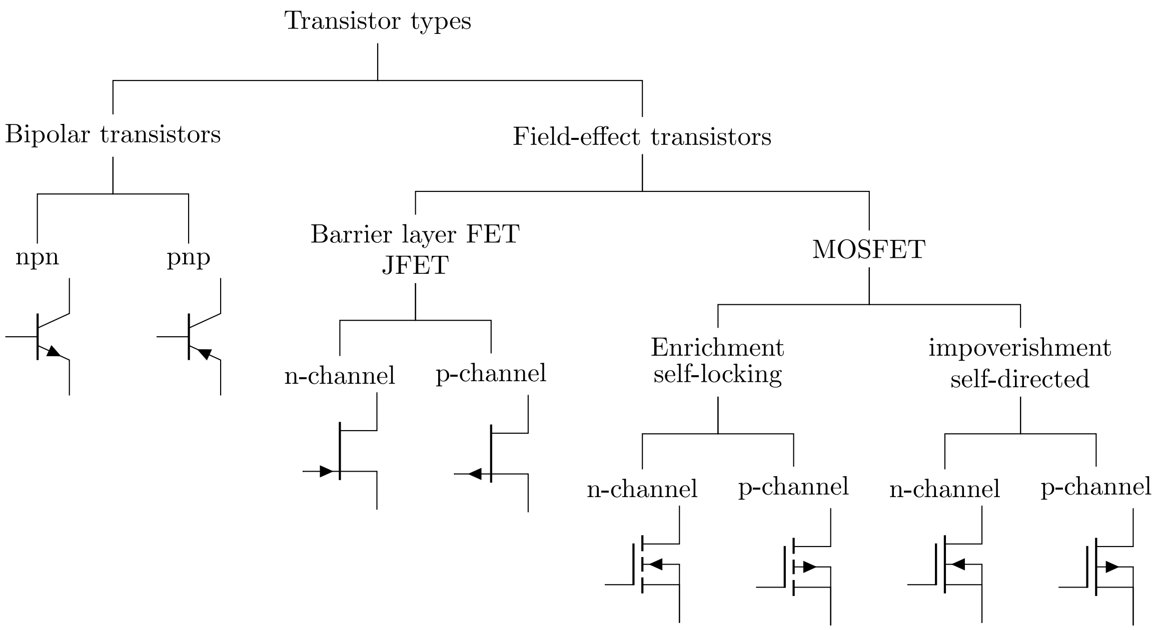


Figure 2.27: **Transistor types.** Overview of the most common types of transistors, both bipolar transistors and unipolar field-effect transistors (FETs). With FETs, a distinction is made between JFETs and MOSFETs (self-blocking and self-conducting), whereby a distinction is made between n-channel and p-channel for each type.

2.3.1 Electrical behaviour

The general electrical behaviour of FETs differs greatly from that of the bipolar transistors discussed so far, in which a base current controls the load current. This requires less power than a bipolar transistor. In FETs, depending on their design, the resistance of the current path is controlled by a voltage applied to the gate and the resulting electric field. The current flow is thus controlled without power.

The following section takes a closer look at the functioning of a self-blocking n-channel MOSFET as an example. The necessary voltages are not shown in relation to the source, but to the fourth connection, bulk (B). In transistors, this connection is usually already connected to the source, which is why they can be equated. For the sake of understanding and the resulting electric fields, it is listed separately.

1. Without applied gate voltage ($U_{\mathrm{GS}} = 0\,\mathrm{V}$)
If there is no voltage between the gate and source (U_{GS}), space charge zones, also known as depletion zones, form at the pn junctions. There are no free charge carriers in the channel area between the drain and source, which is why no current flow is possible.

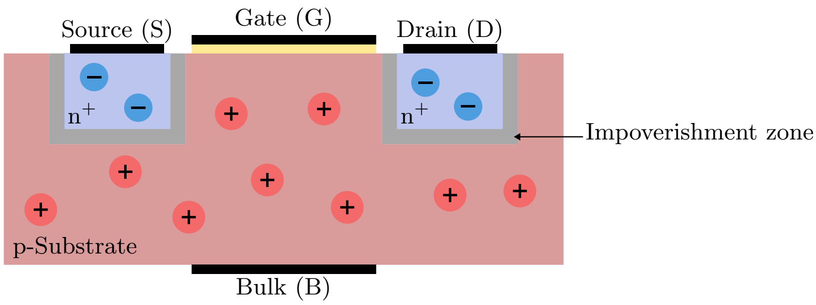


Figure 2.28: **Blocking region n-channel MOSFET.** Cross-section of an n-channel MOSFET (self-blocking) without applied voltages.

2. Positive gate voltage ($U_{\mathrm{GS}} > U_{\mathrm{th}}$)
If a positive voltage greater than the threshold voltage(U_{th}) is applied between the gate and source, the electric field pulls electrons from the p-doped substrate into the vicinity of the gate oxide, leading to the formation of a conductive channel. The so-called inversion zone contains an enrichment of free charge carriers with the opposite sign to the charge carriers that primarily predominate in the semiconductor layer.

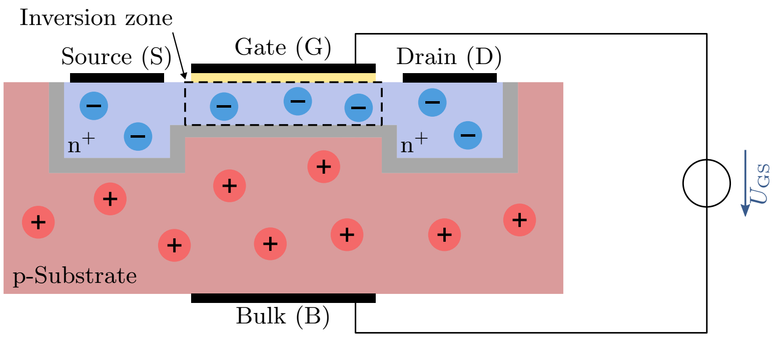


Figure 2.29: **Ohmic region n-channel MOSFET.** Cross-section of an n-channel MOSFET (self-blocking) with applied gate-source voltage greater than the threshold voltage.

3. Electric current ($U_{\mathrm{DS}} > 0\,\mathrm{V}$)
Once an n-channel has been formed, a voltage can be applied between the drain and source (U_{DS}) to generate a current flow. The electrons move from the source to the drain, creating a current flow (I_{D}). At low U_{DS}, the MOSFET is in the linear region, where the current flow I_{D} is proportional to U_{DS}. The behaviour is similar to that of a resistor. At higher U_{DS}, the MOSFET reaches the saturation region, in which I_{D} becomes largely independent of further increases in U_{DS}. The current flow is primarily controlled by U_{GS}.

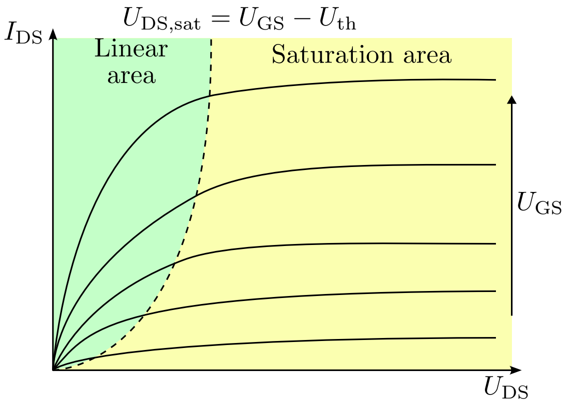


Figure 2.30: **Output characteristic curve field n-channel MOSFET.** Output characteristic curve of an n-channel MOSFET (self-blocking) with the linear region and saturation region. Drain current as a result of the drain-source voltage at different gate-source voltages.

Saturation is caused by the fact that, at high voltage, the free charge carriers are displaced from the channel and the channel is constricted. The charge carriers can still pass through this area, but no further increase in I_D is possible due to U_{DS}.

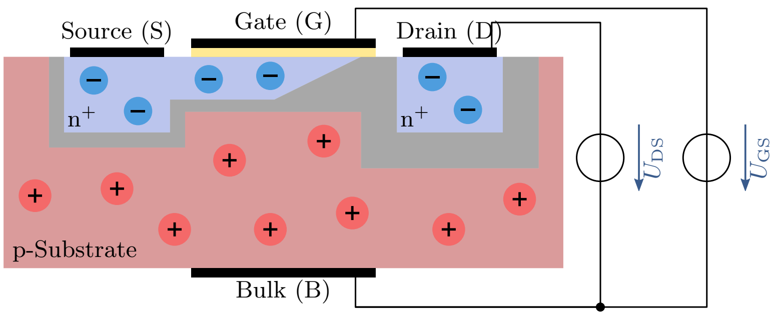


Figure 2.31: **Saturation region n-channel MOSFET.** Cross-section of an n-channel MOSFET (self-blocking) with applied voltage U_{GS} and U_{DS}.

Key point:

- MOSFETs have three terminals: gate (G), source (S) and drain (D).
- The electrical behaviour is controlled without power.
- U_{GS} causes an inversion of the charge carriers below the gate.

After a detailed examination of the self-locking n-channel MOSFET, the following table provides a summary of other types of field-effect transistors. The overview shows the input characteristic curve with I_D as a function of U_{GS} and the output characteristic field with I_D as a function of U_{DS} at different values of U_{GS} are shown. The individual curves differ in the signs of the currents and voltages, depending on the respective doping. In addition, the different types differ in the size of the threshold voltage U_{th}.

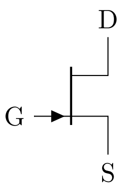
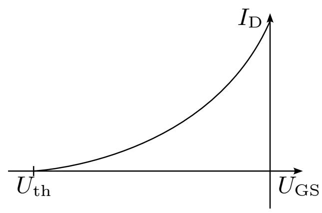
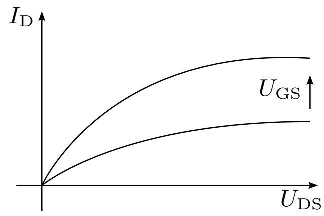
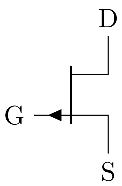
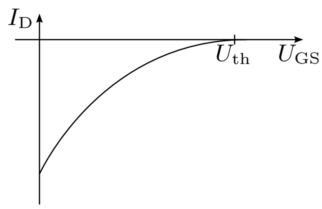
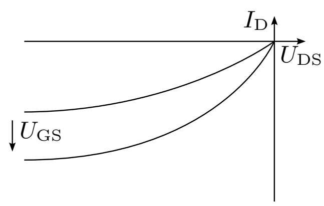
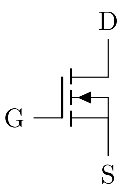
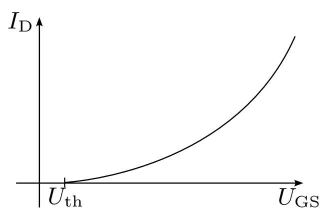
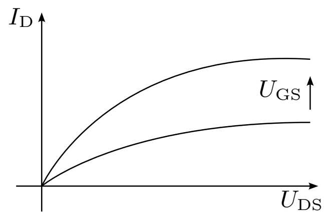
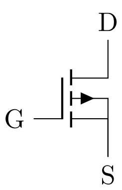
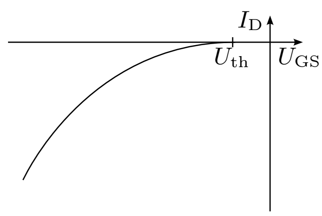
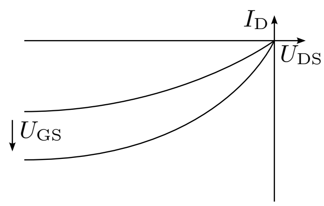
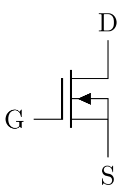
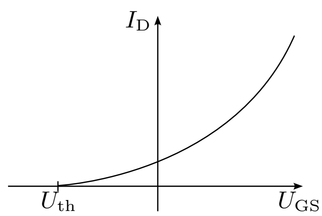
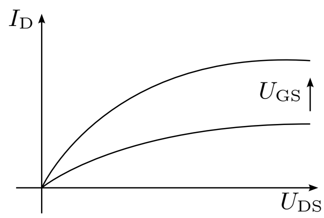
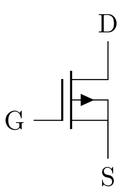
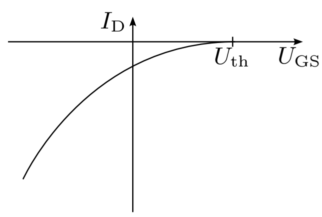
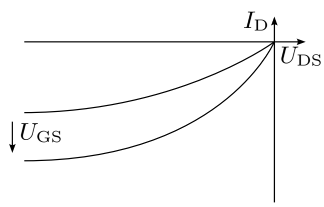
Type	**Symbole**	**Input characteristic curve**	**Output characteristic curve field**
n-channel JFET			
p-channel JFET			
n-channel MOSFET self-locking			
p-channel MOSFET self-locking			
n-channel MOSFET self-directed			
p-channel MOSFET self-directed			

Table 2.3: **Overview of field effect transistors.** Electrical behaviour (input characteristic curve and output characteristic curve field) of various field-effect transistors.

Key point:

- The channel's bias determines the necessary polarity of the currents and voltages at the transistor.
- The input characteristics can be distinguished by the position of the threshold voltage U_{th}.

2.3.2 Structure

The previous section showed a schematic cross-section of an n-channel MOSFET. For many applications and logic circuits, p-channel MOSFETs are also required. Both complementary variants can be realised using the same starting material, which is then referred to as CMOS technology (Complementary Metal-Oxide-Semiconductor). In the first step of the manufacturing process, local n-doping must be carried out for the p-channel MOSFET (PMOS); the rest of the structure corresponds to that of an n-channel MOSFET (NMOS) with complementary doping. Although the M in MOSFET originally stands for metal, for manufacturing reasons the gate material today is usually made of conductive polysilicon. The electrodes and the interconnection of the areas and the gate are, for example, made using aluminium structures.

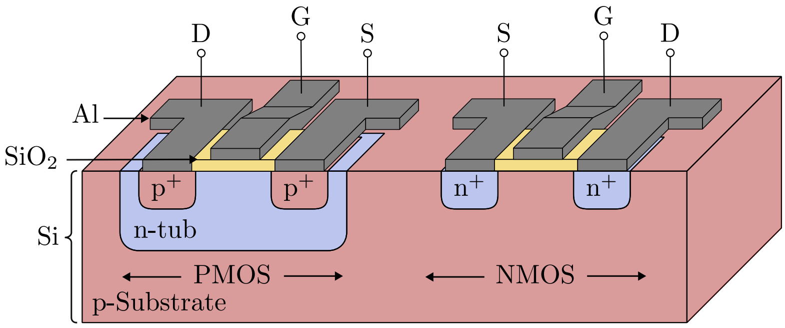


Figure 2.32: **Cross-section CMOS structure.** Integration of a PMOS and NMOS in a common p-doped silicon substrate.

The following figure shows a possible cross-section of an n-channel JFET. In direct comparison to the MOSFET, the structure is significantly simpler due to the absence of the insulation layer on the gate. Although the functioning of a MOSFET was described earlier, the JFET could be manufactured about 15 years earlier due to its simpler structure. The highly doped n^+ areas serve to improve contact with the n-channel and are not absolutely necessary. This means that only three doped areas and their electrical contacts are essential: the n-channel with drain and source connections, and the p-doped gate above and below the channel.

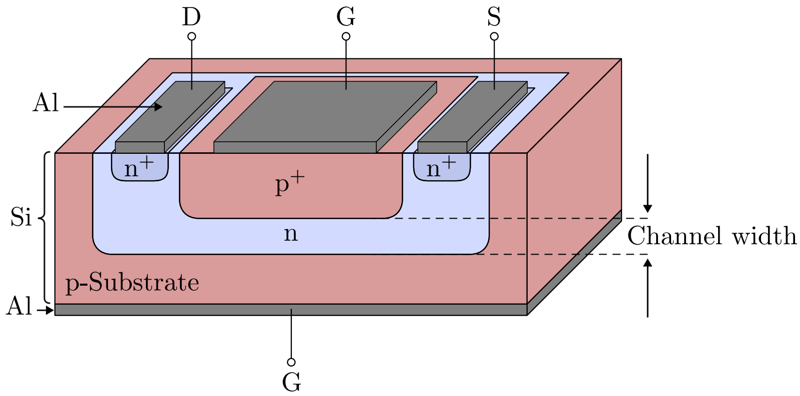


Figure 2.33: **Cross-section of an n-channel JFET.** General structure of an n-channel JFET with doped regions and necessary electrodes.

Key point:

- CMOS technology combines complementary MOSFETs in a single component.
- JFETs have the simplest FET structure.

2.3.3 Applications/Basic circuits

Controllable resistance

When operating in the linear range, the MOSFET has a variable resistance value that can be controlled electronically. The adjustable value can be used as a control element for more complex electronic circuits. An important advantage of using such a transistor is that the control signal is very well isolated from the resistance terminals. The following figure shows how, in the simplest case, a MOSFET serves as a controllable resistor in a voltage divider, as well as a general representation of the circuit represented by it. It should be noted that the circuit shown has only a very limited range of applications. The reasons for this are the limited linearity of the transistor, its low resistance and the influence of the resistor R on U_{DS}.

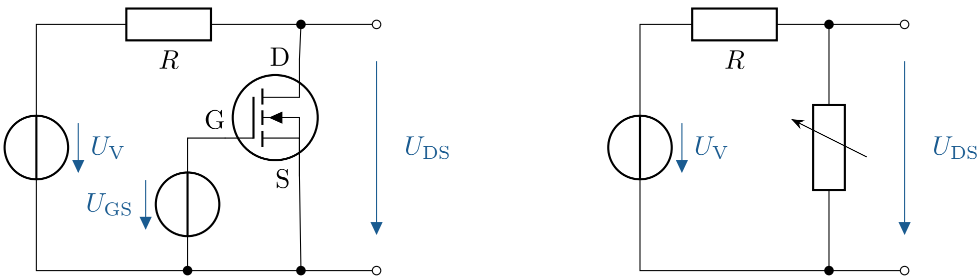


Figure 2.34: **MOSFET as a controllable resistor.** Left: Voltage divider with MOSFET as controllable element. Right: Voltage divider with representation of the MOSFET as controllable resistor.

Key point:

The ohmic resistance of a MOSFET can be controlled by U_{GS}, but the conduction resistance is low and the value range is very narrow.

Switch
A MOSFET can be operated as a switch by controlling the voltage U_{GS}. When switched on, a sufficiently positive voltage (for n-channel MOSFETs) is applied to the gate, creating a conductive channel between the source and drain. This allows current to flow through the transistor and the load connected in series. When switched off, U_{GS} is reduced, which removes the conductive channel and stops the current flow.

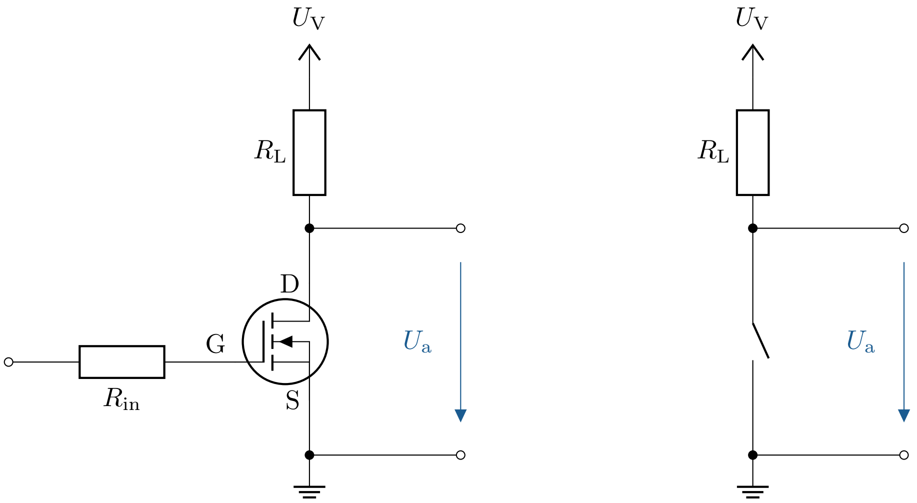


Figure 2.35: **MOSFET as a switch.** Left: Circuit with MOSFET as a controllable switch. Right: Equivalent circuit diagram with the MOSFET represented as a switch.

Key point:

The major advantage of MOSFETs in this scenario is their high switching speed, low resistance in the conductive state and high input resistance at the gate.

Typical applications for MOSFETs as switches are switching power supplies, motor controls, logic circuits and generally as replacements for relays. The operating range is in the saturationrange.

Inverter
As already described in the section on structure, CMOS structures contain both p-channel and n-channel MOS transistors. A typical basic element based on this is an inverter, also known as a NOT gate. The input signal E is fed to the gates connected in parallel. The potential between the drain-source sections connected in series serves as the output signal. The circuit is shown in Figure 2.36 on the left. This results in the following two states:

1. Input signal 0 V (low):

- $Transistor_2$ (T_2) is conductive, since U_{GS2} is negative.
- $Transistor_1$ (T_1) is blocking, since U_{GS1} is zero.
- The current can flow from the supply voltage through T_2 to the output, or 5 V is present at Q (high signal)..

2. Input signal 5 V (high):

- T_2 is blocking, since U_{GS2} is zero.

- T_1 is conductive, since U_{GS1} is positive.
- The current can flow from the output through T_1 to ground, or 0 V is applied to Q (low signal).

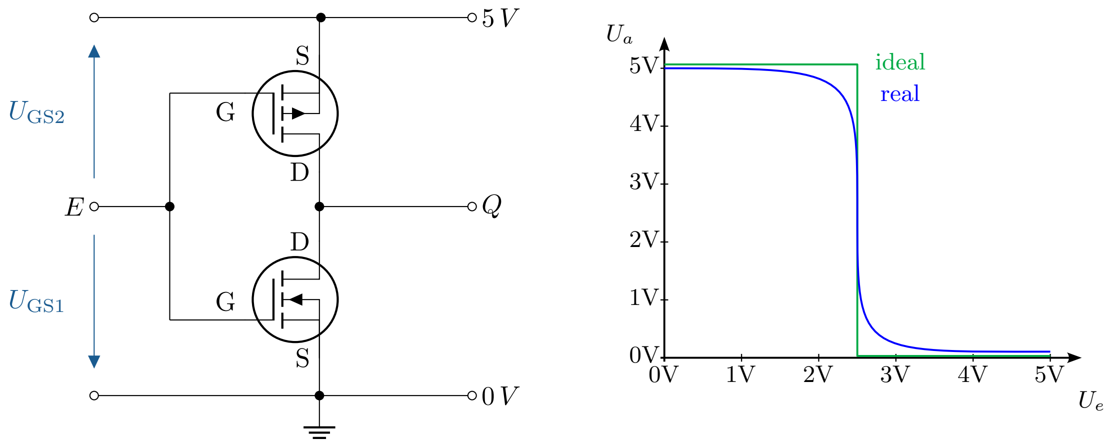


Figure 2.36: **MOSFET as an inverter.** Left: Circuit of a CMOS inverter. Right: Voltage characteristic curve with real output signal (blue) and ideal output signal (green).

According to the behaviour described, a discretised output signal results that is inverse to the input signal (see Figure 2.36 on the right). It should be noted that the desired behaviour only occurs reliably at voltages close to 0 V or the supply voltage. At half the supply voltage, for example, the value of the output signal is not precisely defined and the transistors can be either conducting or blocking, or assume a state in between. In addition, this state should be avoided because the supply voltage is short-circuited and a large current flow is possible.

A Übungsaufgaben

A.1 Beschreibung des Bändermodells

Beschreiben Sie ausführlich das Bändermodell von Festkörpern und seine Bedeutung für die Unterscheidung zwischen Leiter, Halbleiter und Isolator. Gehen Sie dabei insbesondere auf die folgenden Aspekte ein:

- relevante Bänder und deren Besetzung mit Ladungsträgern.
- Bedeutung der Bandlücke und wie diese elektrische Eingeschaften von Materialien beeinflusst.
- Einfluss von Temperaturänderungen auf die Leitfähigkeit.

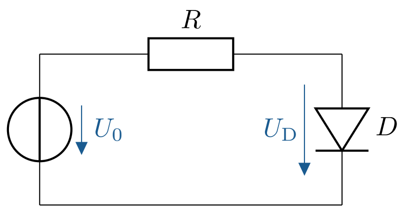
A.2 Die Raumladungszone im pn-Übergang

Untersuchen Sie die Eigenschaften und die Bildung der Raumladungszone in pn-Übergängen von Halbleitern. Beantworten Sie die folgenden Aspekte in Ihrer Ausarbeitung:

- Was ist die Raumladungszone und wie entsteht diese im pn-Übergang?
- Wie ist der Verlauf des elektrischen Feldes im Halbleiter?
- Welche inneren Vorgänge wirken auf die freien Ladungsträger?
- Wie verändert sich die Raumladungszone bei einer angelegten Spannung in Durchlass- und Sperrrichtung?

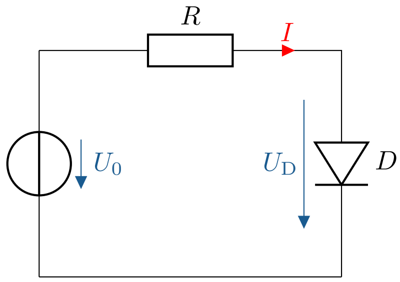
A.3 Berechnung des Spannungsabfalls einer Diode

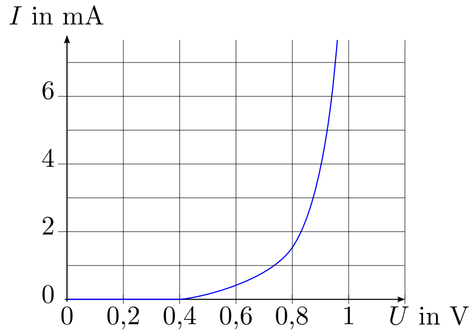
Gegeben ist eine Reihenschaltung aus einer Gleichspannungsquelle, einem Widerstand und einer Siliziumdiode mit einer Schleusenspannung von $U_\mathrm{S} = 0{,}7\,\mathrm{V}$. Welche Spannung fällt näherungsweise über dem Widerstand R ab, wenn die Versorgungsspannung $U_0 = 5\,\mathrm{V}$ beträgt?



A.4 Bestimmung des Arbeitspunkts und der Leistung einer Diode

Eine Diode hat die folgende Kennlinie im Durchlassbereich. Sie wird in Reihe mit einer idealen Spannungsquelle $U_0 = 1\,\mathrm{V}$ und einem Widerstand mit dem Wert von $R = 200\,\Omega$ geschaltet.





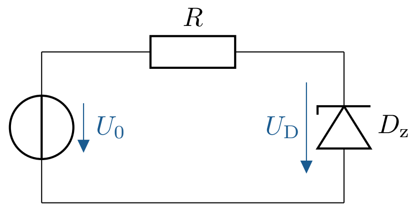
a) Welcher Strom I und welche Spannung U_D stellen sich ein? Lösen Sie das Problem graphisch.

b) Welche elektrische Leistung wird an der Diode umgesetzt?

c) Wie muss R gewählt werden, damit sich ein Strom von $3\,\text{mA}$ einstellt?

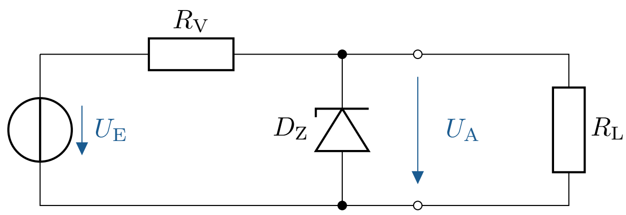
A.5 Dimensionierung des Vorwiderstands einer Zenerdiode

Die Zenerspannung der Diode beträgt $U_D = 6\,\text{V}$ und die Diode darf eine maximale Verlustleistung von $P_{tot} = 400\,\text{mW}$ nicht überschreiten. Die maximale Eingangsspannung beträgt $U_0 = 12\,\text{V}$.
Wie groß muss der Widerstand R mindestens sein, damit die Diode nicht zerstört wird?



A.6 Einsatz einer Zenerdiode zur Spannungsstabilisierung

Mit Hilfe einer Zenerdiode soll ein Lastwiderstand $R_L = 160\,\Omega$ mit einer stabilisierten Spannung von $U_A = 8\,\text{V}$ versorgt werden.
Dem Datenblatt der Zenderdiode ist zu entnehmen, dass bei einer Zenerspannung von $-8\,\text{V}$ ein Zenerstrom von $-0{,}45\,\text{A}$ fließt.



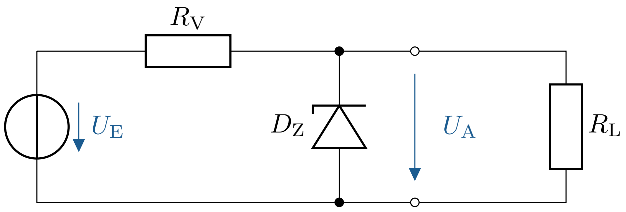
a) Berechnen Sie den Vorwiderstand R_V, sodass sich bei einer Eingangsspannung von $U_E = 10\,\text{V}$ eine Ausgangsspannung von $U_A = 8\,\text{V}$ einstellt.

b) Wie groß sind die Leistungen, die an R_V, R_L und an der Diode D_Z umgesetzt werden?

A.7 Ermittlung der Betriebsgrenzen einer Zenerdiode

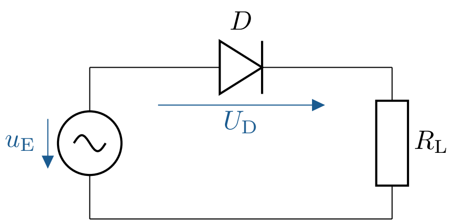
Gegeben ist die folgende Schaltung mit einer Zenerdiode und zwei Widerständen. Die eingesetzte Zenerdiode besitzt eine Zenerspannung von $U_Z = 8\,\text{V}$. Die maximale Verlustleistung der Diode beträgt

$P_{\mathrm{tot}} = 8\,\mathrm{W}$. Die Diode arbeitet im stabilisierenden Bereich, wenn der Strom durch sie mindestens $5\,\mathrm{mA}$ beträgt. Zusätzlich sind die folgenden Werte gegeben: $U_{\mathrm{E}} = 20\,\mathrm{V} \pm 10\%$ und $R_{\mathrm{V}} = 10\,\Omega$.
Welchen minimalen und maximalen Widerstandswert darf der Lastwiderstand R_{L} nicht unter- bzw. überschreiten, damit die Diode einerseits im stabilisierenden Bereich arbeitet und andererseits nicht überlastet wird?



A.8 Einsatz einer Diode zur Leistungsreduzierung

In einer einfachen Schaltung wird eine Diode zur Leistungsreduktion eingesetzt. Der Widerstand $R_{\mathrm{L}} = 1\,\mathrm{k}\Omega$ stellt eine Heizlast dar. Die Netzspannung beträgt 230 V Effektivwert bei einer Frequenz von 50 Hz.

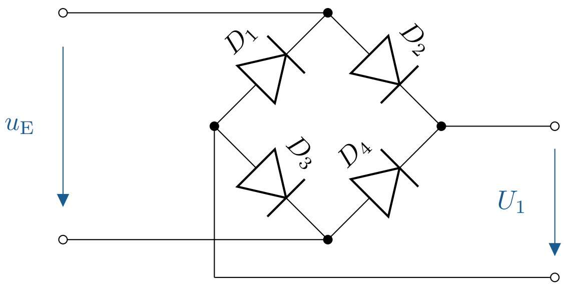


a) Welche mittlere Leistung wird in R_{L} und in der Diode umgesetzt, wenn die Diode als idealer Gleichtrichter wirkt (d.h. nur eine Halbwelle durchlässt)?

b) Wie groß wäre die mittlere Leistung bei überbrückter Diode (volle Netzspannung an R_{L})?

A.9 Spannungsverlauf in einer Brückengleichrichterschaltung

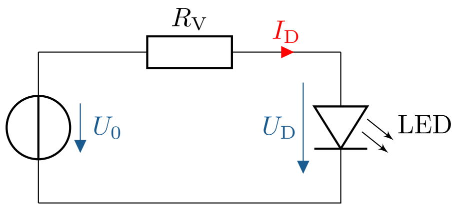
Skizzieren Sie die Spannung U_1 der gegebenen Schaltung. Beachten Sie, dass es sich bei u_{E} um eine Wechselspannung handelt.



A.10 Berechnung des Vorwiderstands einer LED

Wie bei allen Dioden muss auch bei Leuchtdioden (LEDs) der Strom begrenzt werden. Dazu wird ein Vorwiderstand in Reihe geschaltet, wie in der unten gezeigten Schaltung.

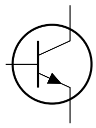
Die eingesetzte LED besitzt eine Durchlassspannung von $U_{\mathrm{D}} = 2{,}2\,\mathrm{V}$. Die ideale Helligkeit wird bei einem Strom von $I_{\mathrm{D}} = 30\,\mathrm{mA}$ erreicht. Die Eingangsspannung beträgt $U_0 = 5\,\mathrm{V}$.



Welchen der folgenden Widerstandswerte würden Sie für R verwenden, um die LED möglichst nahe an ihrem optimalen Betriebspunkt (d.h. mit optimaler Helligkeit) zu betreiben?
$15\,\Omega$, $33\,\Omega$, $68\,\Omega$, $150\,\Omega$, $180\,\Omega$

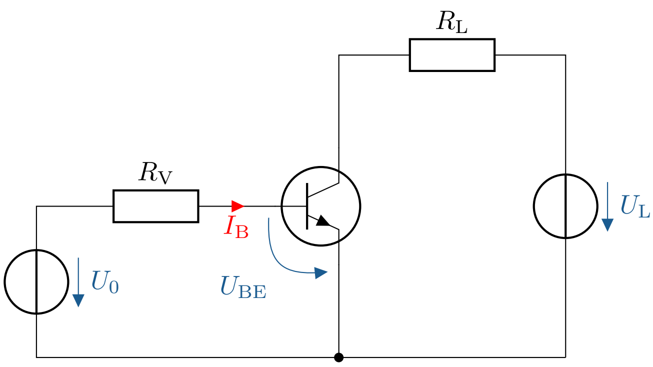
A.11 Identifikation eines Bauteils anhand des Schaltzeichens

Was für ein Bauelement wird mit folgendem Symbol gekennzeichnet?



A.12 Ermittlung von Basis-Emitter-Spannung und Basisstrom

Gegeben ist die abgebildete Transistorschaltung mit den folgenden Parametern: $U_0 = 5\,\mathrm{V}$, $U_L = 15\,\mathrm{V}$, $R_{\mathrm{V}} = 100\,\mathrm{k}\Omega$ und $R_{\mathrm{V}} = 500\,\Omega$.

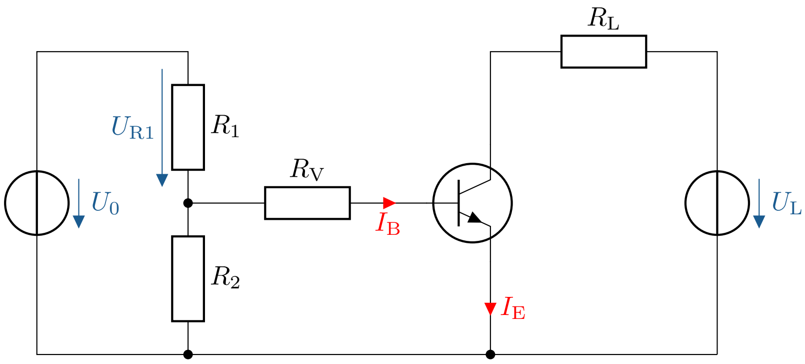


a) Wie groß ist die Spannung U_{BE} näherungsweise, wenn es sich um einen Siliziumtransistor handelt?

b) Wie groß ist der Strom I_{B}?

A.13 Berechnung der Widerstände einer Transistorschaltung

Gegeben ist die abgebildete Transistorschaltung mit den folgenden Parametern: $U_0 = 5\,\mathrm{V}$, $U_{\mathrm{R1}} = 0{,}6\,\mathrm{V}$, $U_{\mathrm{L}} = 15\,\mathrm{V}$, $R_1 = 300\,\Omega$, $R_2 = 2\,\mathrm{k}\Omega$ und $R_{\mathrm{V}} = 500\,\Omega$. Damit der gezeichnete Siliziumtransistor schaltet, braucht er einen Basisstrom von $50\,\mu\mathrm{A}$ um durchzuschalten.

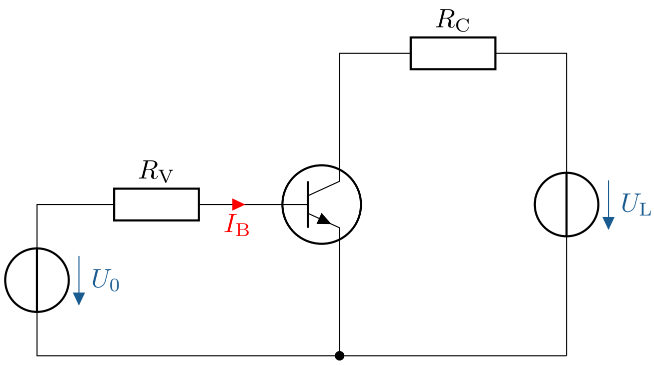


a) Wie groß muss der Widerstand R_V gewählt werden, damit der Transistor durchschaltet?

b) Angenommen der Spannungsabfall des Transistors U_{CE} beträgt 7,5 V, wie groß ist der Strom durch I_E?

A.14 Schaltverhalten eines Transistors

Gegeben ist die abgebildete Transistorschaltung mit den folgenden Parametern: $U_0 = 5\,\text{V}$ und $U_L = 15\,\text{V}$. Damit der gezeigte Siliziumtransistor schaltet, benötigt er einen Basisstrom von $I_B = 50\,\mu\text{A}$, um in den leitenden Zustand zu gelangen.

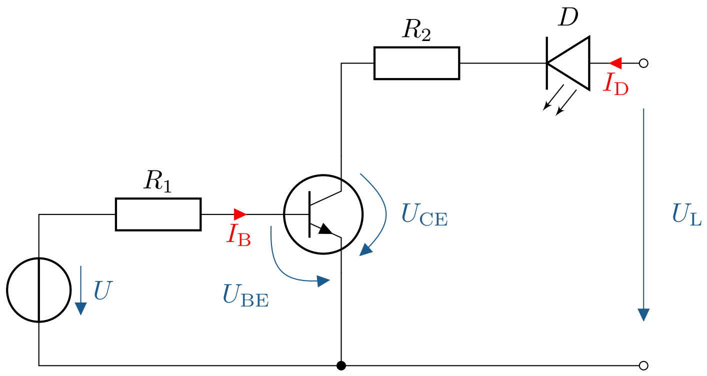


a) Wie groß muss der Vorwiderstand R_V gewählt werden, damit der Transistor durchschaltet?

b) Der Transistor hat einen Stromverstärkungsfaktor $\beta = 180$ und einen Kollektorwiderstand $R_C = 200\,\Omega$. Wie groß sind der Kollektorstrom I_C und der Spannungsabfall zwischen Kollektor und Emitter U_{CE}?

A.15 Transistorschaltung mit LED

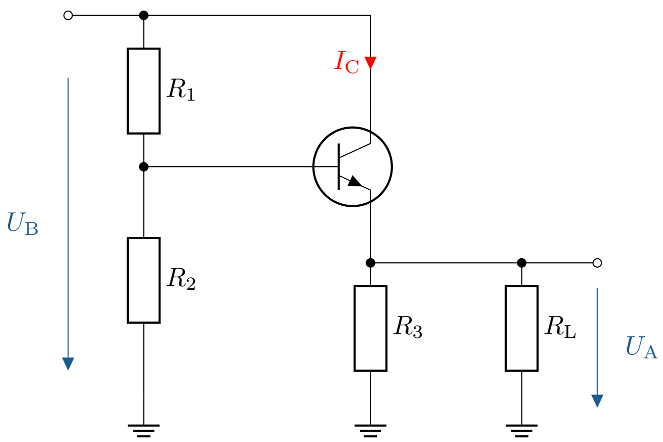
Mithilfe eines Bipolartransistor soll eine LED betrieben werden. Dabei muss durch die LED ein Strom von $I_D = 30\,\text{mA}$ fließen, damit sie ihre optimale Leuchtkraft erreicht.
Der verwendete Transistor besitzt einen Stromverstärkungsfaktor von $\beta = 50$ und im durchgeschalteten Zustand eine Kollektor-Emitter-Spannung von $U_{CE} = 0{,}3\,\text{V}$. Zusätzlich sind die folgenden Parameter gegeben: $R_1 = 3{,}3\,\text{k}\Omega$, $R_2 = 270\,\Omega$ und $U_L = 9\,\text{V}$.



Welche Spannung U muss an der Basis anliegen, damit der Transistor durchschaltet?

A.16 Analyse des Arbeitsbereichs eines Transistors

Der Stromverstärkungsfaktor des verwendeten Siliziumtransistors beträgt $\beta = 150$ und die Versorgungsspannung liegt bei $U_V = 15\,\text{V}$. Die Schaltung soll so dimensioniert werden, dass der Kollektorstrom $I_C = 2\,\text{mA}$ beträgt.

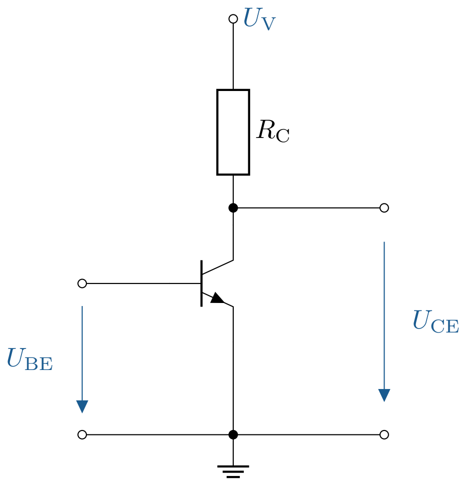


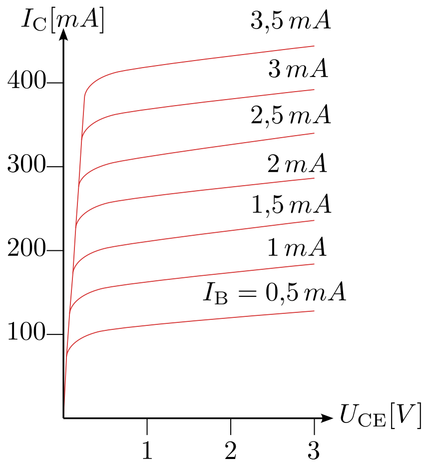
a) Dimensionieren Sie den Widerstand R_3 so, dass bei einer Spannung am Ausgang von $U_A = 3\,\text{V}$ der Strom durch R_3 genau 75% des Emitterstroms beträgt.

b) Bestimmen Sie die Widerstände R_1 und R_2, unter der Annahme, dass die Kollektor-Basis-Spannung $U_{BE} = 0{,}6\,\text{V}$ beträgt und durch R_2 etwa das 9-fache des Basisstroms I_B fließt.

A.17 Berechnungen in einer Transistorschaltungen

Gegeben ist die folgende Schaltung mit dem Ausgangskennlinienfeld eines Transistors.





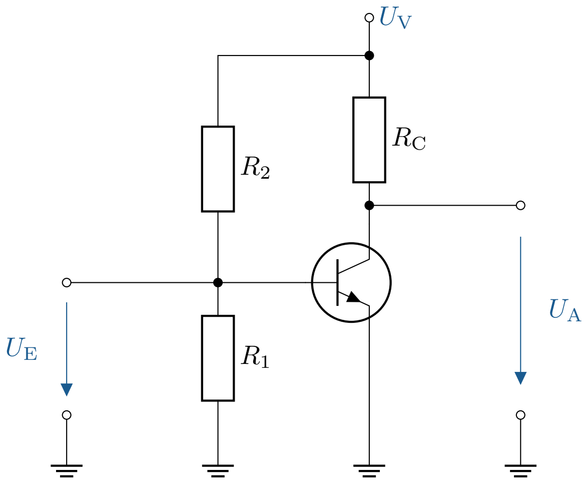
a) Wie groß ist der maximale Kollektorstrom des idealen Transistors (bei $R_{CE} = 0\,\Omega$), wenn der Kollektorwiderstand $R_C = 7{,}5\,\Omega$ beträgt und die Versorgungsspannung $U_V = 3\,V$ ist?

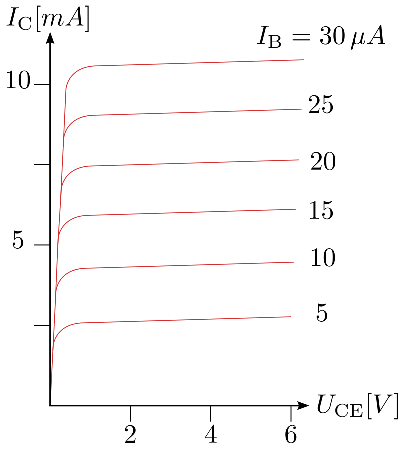
b) Bestimmen Sie U_{CE}, U_{RC} und I_C des realen Transistors bei einem Basisstrom von $I_B = 2\,mA$. Tragen Sie dazu die Widerstandsgerade in das Kennlinienfeld ein.

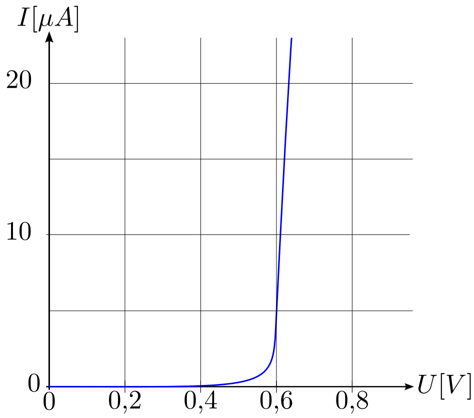
c) Für einen neuen Arbeitspunkt fallen nur noch 1,7 V über dem Kollektorwiderstand R_C ab, der Basisstrom beträgt weiterhin 2 mA. Bestimmen Sie den neuen Widerstand R_C.

A.18 Berechnung der Stromverstärkung einer Transistorschaltung

Gegeben ist die folgende Schaltung mit dem Kennlinienfeld des Transistors. Die Versorgungsspannung beträgt $U_V = 6\,V$. Der Arbeitspunkt liegt bei der halben Betriebsspannung.





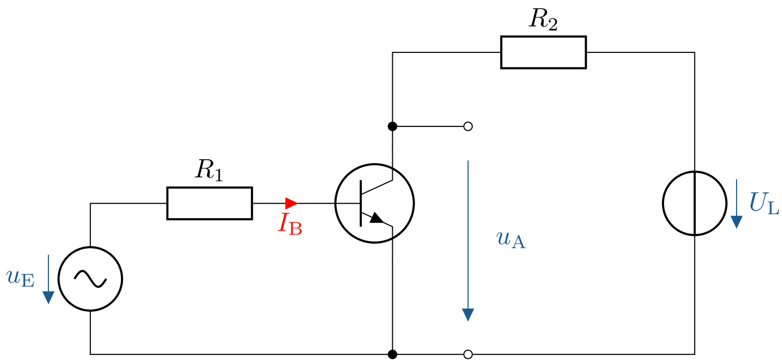


Bestimmen Sie die folgenden Werte aus dem Kennlinienfeld:

- Die Paare I_B/U_{BE} und I_C/U_{CE} im Arbeitspunkt
- Die Stromverstärkung β

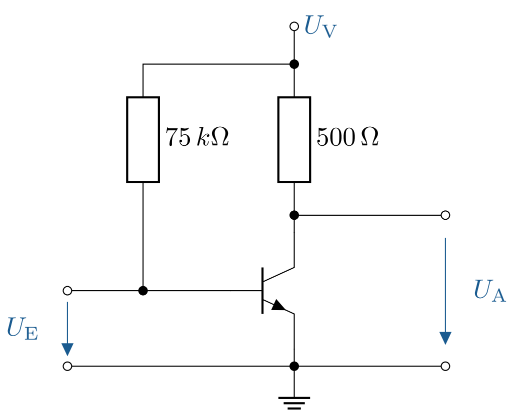
A.19 Verhalten eines Transistors bei Wechselspannung

Was passiert wenn an einen Transistor eine reine Wechselspannung angelegt wird?



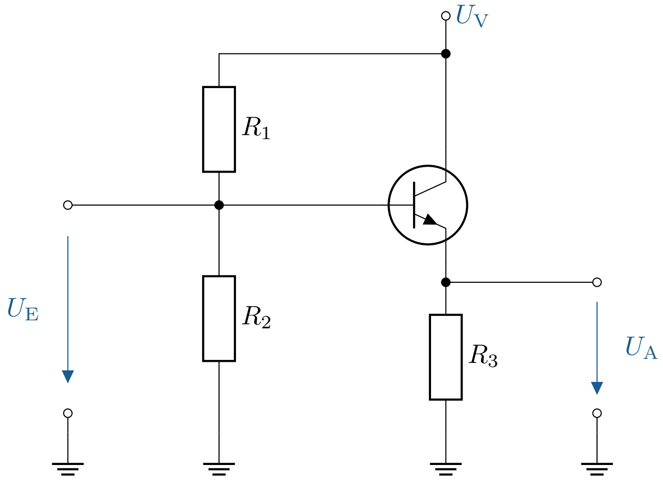
A.20 Erstellung eines Kleinsignal-Ersatzschaltbilds

Zeichnen Sie das Kleinsignalersatzschaltbild der gegebenen Schaltung.



A.21 Erstellung eines Kleinsignal-Ersatzschaltbilds

Zeichnen Sie das Kleinsignalersatzschaltbild der gegebenen Schaltung.

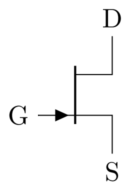


A.22 Bestimmung der Gate-Source-Spannung eines Feldeffekttransistors

Bei welcher Spannung an Source-Gate leitet ein Sperrschichttransistor den meisten Strom?

A.23 Untersuchung des Stromflusses im Feldeffekttransistor (FET)

Wo fließt bei dem dargestellten Transistor der meiste Strom?

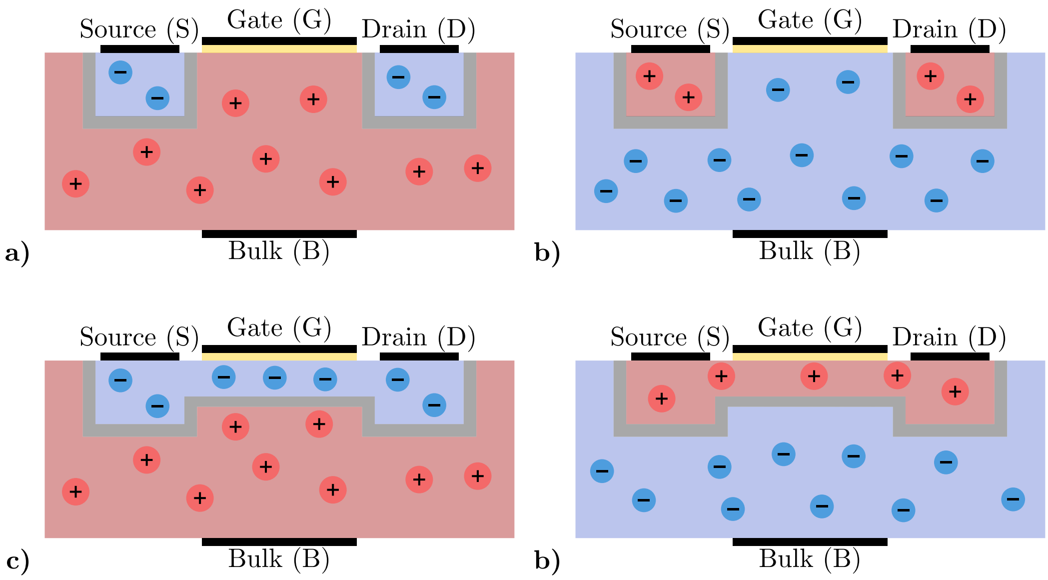


A.24 Interpretation der Kennlinie eines Sperrschichttransistors

In welche zwei Bereiche lässt sich die Ausgangskennlinie des Sperrschichttransistors unterteilen?

A.25 Erkennung von Dotierungsmustern bei Transistoren

Ohne das Anlegen einer Spannung ergeben sich die folgenden Verteilungen von p- und n-Gebieten. Um was für einen Typ von Transistor handelt es sich?



A.26 Erklärung des Funktionsprinzips der CMOS-Technologie

Was sagt das C in CMOS aus?

a) Unterscheidung von Anreicherungs- und Verarmungstyp

b) Die Verwendung von JFETs und MOSFETS in einer Schaltung

c) Spiegelt den Verlauf der Transferkennlinie wieder

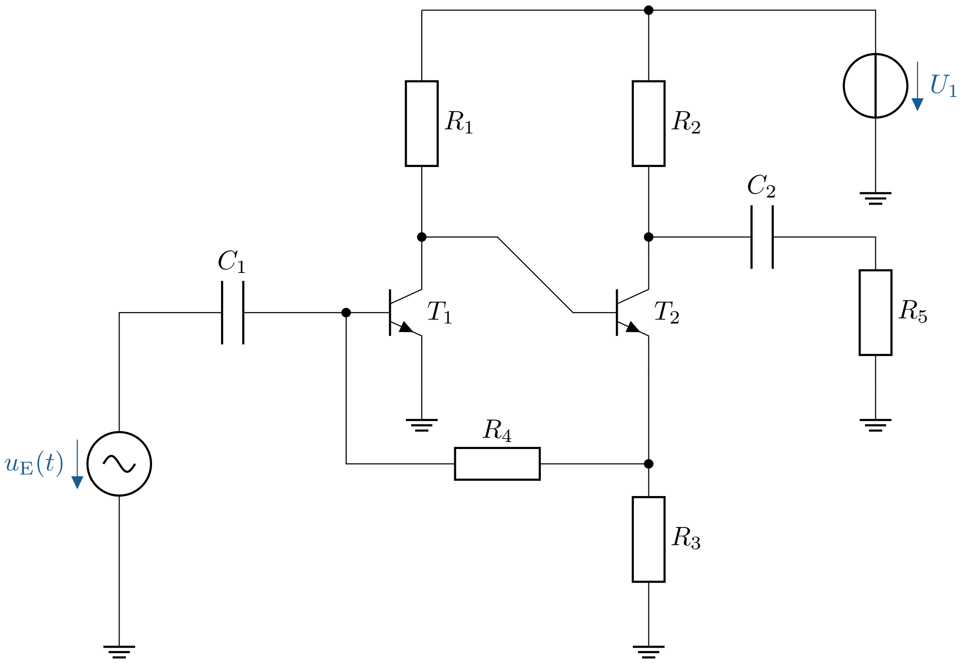
d) Weißt auf die Spannungsempfindlichkeit von MOSFETs hin

e) In einer Schaltung werden sowohl p- als auch n-Kanal Transistoren verwendet

A.27 Transistor-Verbundverstärker

Zwei npn-Bipolartransistoren bilden zusammen einen rauscharmen Verstärker für Signale im Bereich unter $1\,\mu$V. Die Versorgungsspannung beträgt $U_1 = 10\,$V. Am Kollektor des Transistors T_2 liegt eine Spannung von $4\,$V gegenüber Masse an. Im Arbeitspunkt von T_1 und T_2 kann für die Basis-Emitter-Spannungen jeweils $U_{\mathrm{BE}} = 600\,$mV angenommen werden. Die Kollektorströme I_{C} sind sehr groß gegenüber den Basisströmen I_{B} anzunehmen, daher gilt näherungsweise $I_{\mathrm{C}} = I_{\mathrm{E}}$. Zusätzlich sind die folgenden Parameter gegeben:

$R_1 = 10\,\mathrm{k\Omega}$, $R_2 = 10\,\mathrm{k\Omega}$, $R_3 = 2\,\mathrm{k\Omega}$, $R_4 = 300\,\mathrm{k\Omega}$, $R_5 = 10\,\mathrm{M\Omega}$
$C_1 = 1\,\mu$F, $C_2 = 1\,\mu$F
$U_1 = 10\,$V, $\hat{u}_{\mathrm{E}} = 0{,}1\,$mV



a) Wie groß ist der maximale Strom durch R_2 im Arbeitspunkt ($I_{\mathrm{R2}} = I_{\mathrm{C,\,T2}}$)?

b) Wie groß ist der Strom durch R_3 näherungsweise, wenn $R_4 >> R_3$?

c) Wie groß ist die Spannung, die über R_3 abfällt?

d) Wie groß ist die Gleichspannung an der Basis von T_2 ($U_{\mathrm{B,\,T2}} = U_{\mathrm{CE,\,T1}}$)?

e) Wie groß ist der Strom durch R_1 ($I_{\mathrm{R1}} = I_{\mathrm{C,\,T1}}$)?

f) Erläutern Sie die Arbeitspunktstabiliserung der gesamten Schaltung auf Grund der Gegenkopplung über R_4.

g) Wie groß ist die Steilheit S von T_1, wenn zusätzlich $U_{\mathrm{T}} = 26\,\mathrm{mV}$ gegeben ist?

h) Wie groß ist die Spannungsverstärkung $A_{\mathrm{V}} = \frac{\Delta U_{\mathrm{CE}}}{\Delta U_{\mathrm{BE}}}$, der ersten Transistorstufe?

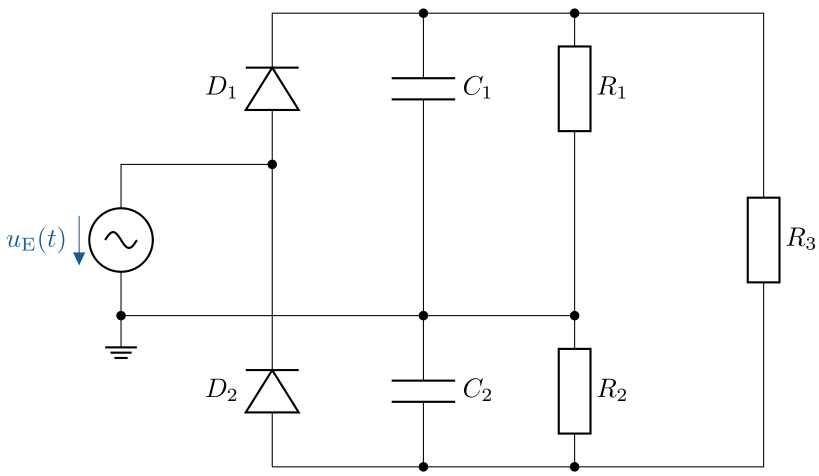
i) Wie groß ist die Spannungsverstärkung A_{V} der zweiten Transistorstufe, unter Verwendung der Definition $A_{\mathrm{V,T2}} = \frac{\Delta U_{\mathrm{C}}}{\Delta U_{\mathrm{B}}}$? Zur Lösung müssen zusätzlich die Widerstände R_2 und R_3 berücksichtigt werden.

j) Wie groß ist die Spannungsverstärkung der beiden Stufen T_1 und T_2 zusammen?

k) Zeichnen Sie das Kleinsignal-Ersatzschaltbild der gesamten Schaltung. Ersetzen Sie darin die beiden Transistoren im Arbeitspunkt jeweils durch eine steuerbare Stromquelle $S = \beta \cdot i_{\mathrm{B}}$ und einen differentiellen Eingangswiderstand r_{BE}.

A.28 Gleichrichterschaltung

Hier eine Gleichrichterschaltung mit pn-Silizium-Dioden. Die Spannungsquelle V_1 liefert sinusförmige 50 Hz Wechselspannung mit einer Amplitude von 5 V.



a) Welche Gleichspannungen treten über R_1,R_2 und R_3 auf?

b) Welche Funktionen haben die Kondensatoren C_1 und C_2?

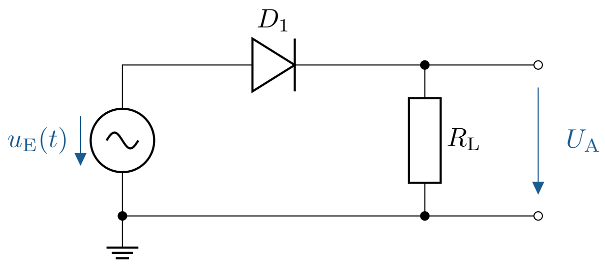
c) Was ist der besondere Vorteil dieser Gleichrichterschaltung, z.B. für Anwendungen mit Operationsverstärkern?

d) Skizzieren Sie eine andere Spannungs-Verdopplungs-Schaltung, bei der die maximale positive Gleichsspannung (Ausgangsspannung) gegenüber Massepotential der speisenden Wechselspannung V_1 zur Verfügung steht.

A.29 Untersuchung eines einfachen Gleichrichters

Analysieren Sie den folgenden Gleichrichter. Zusätzlich sind die folgenden Parameter gegeben:

$\hat{u}_E = 5\,\mathrm{V}$, $R_L = 1\,\mathrm{k\Omega}$ und eine Diodendurchlassspannung von $U_f = 0{,}7\,\mathrm{V}$.



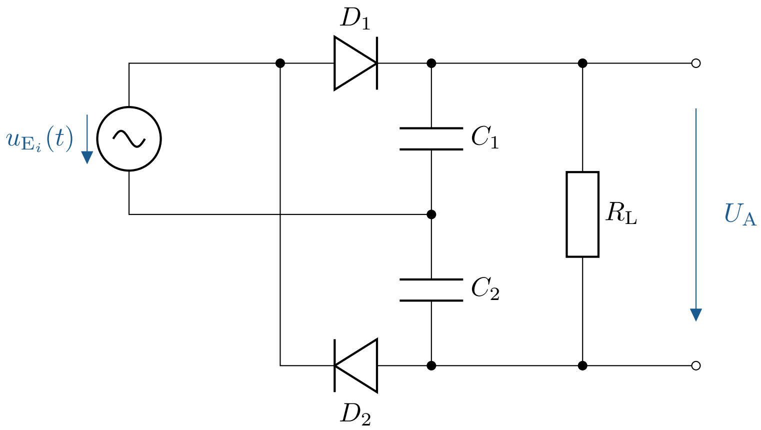
a) Berechnen Sie die Ausgangsspannung U_A.

b) Bestimmen Sie den durch den Lastwiderstand R_L fließenden Strom I_L.

A.30 Analyse einer Spannungsverdoppler-Schaltung

Analysieren Sie die Funktion der dargestellten Delon-Schaltung zur Spannungsverdopplung und berechnen Sie die Ausgangsspannung für verschiedene Eingangsspannungen. Zusätzlich sind die folgenden Parameter gegeben:

$U_{E1,\mathrm{eff}} = 110\,\mathrm{V}$, $U_{E2,\mathrm{eff}} = 230\,\mathrm{V}$ und der Diodendurchlassspannung von $U_f = 0{,}7\,\mathrm{V}$.



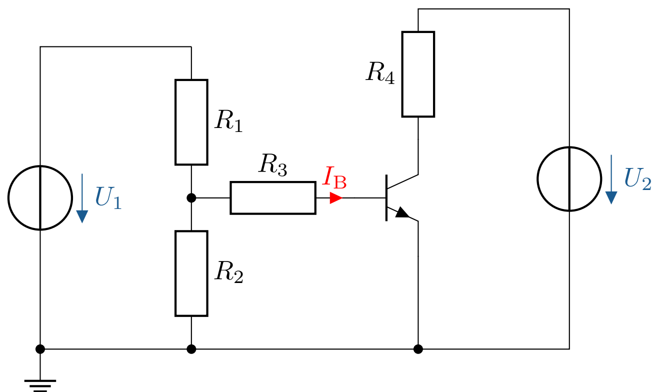
a) Erklären Sie das Funktionsprinzip der Delon-Schaltung und wie diese die Eingangsspannung verdoppelt.

b) Berechnen Sie die theoretische Leerlauf-Ausgangsspannung U_A für die gegebenen Eingangsspannungen.

c) Diskutieren Sie, wie sich die Ausgangsspannung U_A unter Lastbedingungen verhält und welche Faktoren die Spannungsstabilität beeinflussen.

A.31 Grundverhalten eines npn-Bipolartransistors

Analysieren Sie das Grundverhalten eines Bipolartransistors in einer Basisschaltung. Der npn-Transistor weist einen Stromverstärkungsfaktor von $\beta = 100$ auf, es wird ein Basisstrom von $I_B = 20\,\mu A$ angenommen.



a) Berechnen Sie den Kollektorstrom I_C.

b) Bestimmen Sie die Kollektor-Emitter-Spannung U_{CE}, wenn der Kollektorwiderstand $R_4 = 1\,k\Omega$ und die Versorgungsspannung $U_2 = 12\,V$ beträgt.

A.32 Zeichnung und Berechnung einer Kollektorschaltung

a) Zeichnen Sie die folgende Schaltung: Beschalten Sie einen npn-Transistor mit einem Basisvorwiderstand von $R_B = 4{,}7\,k\Omega$, einem Emitterwiderstand von $R_E = 1\,k\Omega$ und einer Spannungsversorgung von $U_0 = 15\,V$. Schalten Sie eine 10 V-Z-Diode in Sperrrichtung von der Basis auf Masse.

Verwenden Sie nur eine Spannungsversorgung, welche sowohl die Basis als auch den Kollektor versorgt.

b) Berechnen Sie die Emitterspannung U_E, und den Emitterstrom I_E.

c) Berechnen Sie die Leistung am Emitterwiderstand P_RE.

B Lösungen zu den Übungsaufgaben

B.1 Beschreibung des Bändermodells

Das Bändermodell erklärt die Energiezustände, die Elektronen in Festkörpern einnehmen können. In einem einzelnen Atom sind die Elektronenzustände diskrete Energielevel. Siehe: Abschnitt 1.1

B.2 Die Raumladungszone im pn-Übergang

Die RLZ ist eine schmale Region um den pn-Übergang herum, in der positive Ionen aus der n-Schicht und negative Ionen aus der p-Schicht verbleiben, nachdem die Diffusion abgeschlossen ist. In dieser Zone gibt es keine freien Ladungsträger, da die positiven und negativen Ladungen sich gegenseitig neutralisieren. Dadurch entsteht ein elektrisches Feld ($\vec{E}$), das sowohl einen Driftstrom erzeugt als auch die Diffusion von weiteren Ladungsträgern unterdrückt. Die Raumladungszone wirkt wie eine Sperrschicht und verhindert den Stromfluss in Sperrrichtung. Siehe: Abschnitt 1.4

B.3 Berechnung des Spannungsabfalls einer Diode

Da es sich um eine Siliziumdiode handelt, beträgt die Schleusenspannung und der daraus resultierende Spannungsabfall am Bauteil ungefähr 0,6 bis 0,7 V.

In der Reihenschaltung ergibt sich der folgende Zusammenhang:

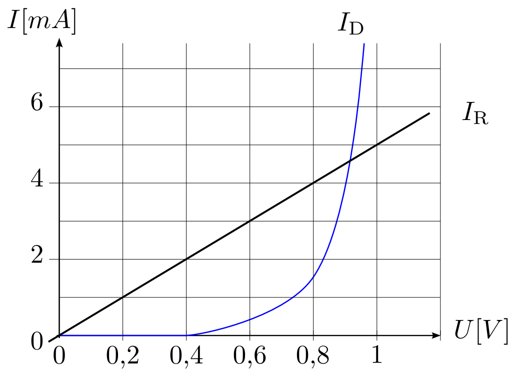
$$U_0 = U_\mathrm{R} + U_\mathrm{D}$$

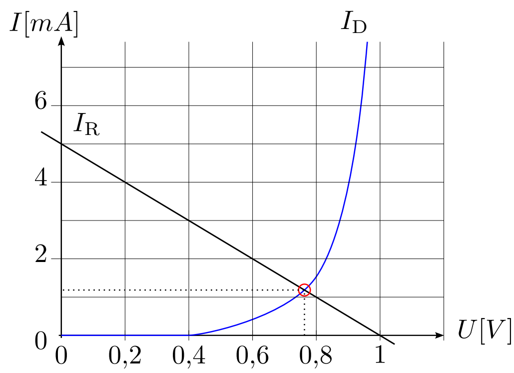
$$U_\mathrm{R} = U_0 - U_\mathrm{D} = 5\,\mathrm{V} - 0{,}7\,\mathrm{V} = 4{,}3\,\mathrm{V}$$

Siehe: Abschnitt 2.1.1

B.4 Bestimmung des Arbeitspunkts und der Leistung einer Diode

a) Im ersten Schritt wird die Wirkung der Spannungsquelle auf den Widerstand R betrachtet. Bei $U_0 = 1\,\mathrm{V}$ fließen durch den Widerstand R ein Strom von $I_\mathrm{R} = \frac{U_0}{R} = \frac{1\,\mathrm{V}}{200\,\Omega} = 5\,\mathrm{mA}$. Es kann die Widerstandgerade vom Ursprung aus zu dem Punkt (1 V, 5 mA) eingezeichnet werden (linke Abbildung).
Gemäß des Aufbaus der Schaltung muss die Widerstandsgerade im nächsten Schritt gespiegelt werden, es entsteht die rechte Darstellung. Der Schnittpunkt beider Linien entspricht dem Arbeitspunkt (U_AP, I_AP), dieser lautet: (0,75 V, 1,2 mA).

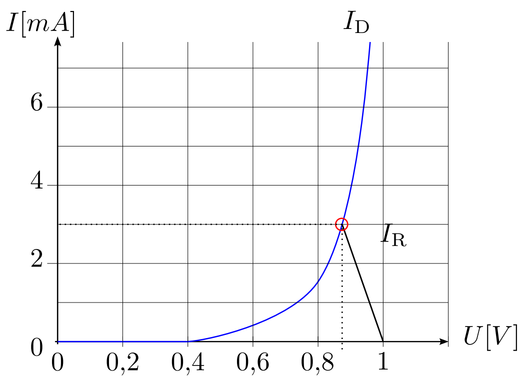




b) Die Leistung ergibt sich aus dem zuvor ermittelten Arbeitspunkt.

$$P = U_{\mathrm{AP}} \cdot I_{\mathrm{AP}} = 0{,}75\,\mathrm{V} \cdot 1{,}2\,\mathrm{mA} = 0{,}9\,\mathrm{mW}$$

c) Ausgehend vom Punkt $(1\,\mathrm{V}, 0\,\mathrm{mA})$ zum Schnittpunkt mit der Diodenkennlinie bei $I = 3\,\mathrm{mA}$ ergibt sich die gesuchte Widerstandsgerade.



Es kann der Schnittpunkt $(0{,}86\,\mathrm{V}, 3\,\mathrm{mA})$ abgelesen werden.

$$U_{\mathrm{R}} = U_0 - U_{\mathrm{D}} = 1\,\mathrm{V} - 0{,}86\,\mathrm{V} = 0{,}14\,\mathrm{V}$$
$$I = 3\,\mathrm{mA}$$
$$R = \frac{U_{\mathrm{R}}}{I} = \frac{0{,}14\,\mathrm{V}}{3\,\mathrm{mA}} = 46{,}67\,\Omega$$

Siehe: Abschnitt 2.2

B.5 Dimensionierung des Vorwiderstands einer Zenerdiode

Die Spannung am Widerstand ergibt sich zu:

$$U_{\mathrm{R}} = U_0 - U_{\mathrm{D}} = 12\,\mathrm{V} - 6\,\mathrm{V} = 6\,\mathrm{V}$$

Die maximale Verlustleistung der Diode ist:

$$P_{\mathrm{tot}} = U_{\mathrm{D}} \cdot I = 400\,\mathrm{mW}$$

Daraus folgt der maximale Strom durch die Diode:

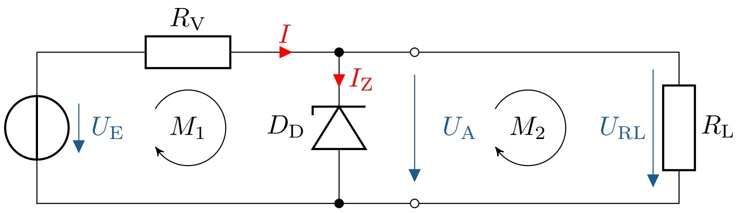
$$I = \frac{400\,\mathrm{mW}}{6\,\mathrm{V}} = 66{,}7\,\mathrm{mA}$$

Um diesen Strom nicht zu überschreiten, ergibt sich der Mindestwert für den Widerstand:

$$R = \frac{U_\mathrm{R}}{I} = \frac{6\,\mathrm{V}}{66{,}7\,\mathrm{mA}} \approx 90\,\Omega$$

Siehe: Abschnitt 2.1.3

B.6 Einsatz einer Zenerdiode zur Spannungsstabilisierung



a)

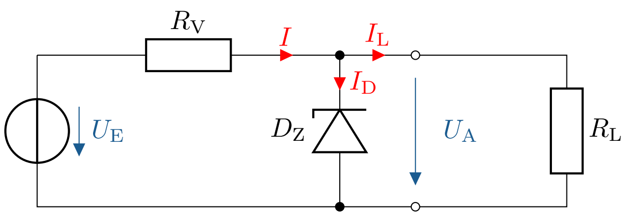
$$I_\mathrm{RL} = \frac{8\,\mathrm{V}}{160\,\Omega} = 0{,}05\,\mathrm{A}$$
$$I = 0{,}05\,\mathrm{A} + 0{,}45\,\mathrm{A} = 0{,}5\,\mathrm{A}$$
$$U_\mathrm{RV} = 10\,\mathrm{V} - 8\,\mathrm{V} = 2\,\mathrm{V}$$
$$R_\mathrm{V} = \frac{2\,\mathrm{V}}{0{,}5\,\mathrm{A}} = 4\,\Omega$$

b)

$$P_\mathrm{RV} = 2\,\mathrm{V} \cdot 0{,}5\,\mathrm{A} = 1\,\mathrm{W}$$
$$P_\mathrm{RL} = 8\,\mathrm{V} \cdot 0{,}05\,\mathrm{A} = 0{,}4\,\mathrm{W}$$
$$P_\mathrm{DZ} = 8\,\mathrm{V} \cdot 0{,}45\,\mathrm{A} = 3{,}6\,\mathrm{W}$$

Siehe: Abschnitt 2.1.3

B.7 Ermittlung der Betriebsgrenzen einer Zenerdiode



Gegeben sind:

$$U_\mathrm{max} = 22\,\mathrm{V} \text{ und } U_\mathrm{min} = 18\,\mathrm{V}$$

Fall 1: Bei $U_{\max} = 22\,\mathrm{V}$

$$
\begin{aligned}
U_{\mathrm{RV}} &= 22\,\mathrm{V} - 8\,\mathrm{V} = 14\,\mathrm{V}\\
I &= \frac{14\,\mathrm{V}}{10\,\Omega} = 1{,}4\,\mathrm{A}\\
I &= I_{\mathrm{D}} + I_{\mathrm{L}}\\
I_{\mathrm{D,\ max}} &= \frac{8\,\mathrm{W}}{8\,\mathrm{V}} = 1\,\mathrm{A}\\
I_{\mathrm{L}} &= 1{,}4\,\mathrm{A} - 1\,\mathrm{A} = 0{,}4\,\mathrm{A}\\
R_{\mathrm{L}} &= \frac{8\,\mathrm{V}}{0{,}4\,\mathrm{A}} = 20\,\Omega \Rightarrow R_{\mathrm{L,\ max}} = 20\,\Omega
\end{aligned}
$$

Fall 2: Bei $U_{\min} = 18\,\mathrm{V}$

$$
\begin{aligned}
U_{\mathrm{RV}} &= 18\,\mathrm{V} - 8\,\mathrm{V} = 10\,\mathrm{V}\\
I &= \frac{10\,\mathrm{V}}{10\,\Omega} = 1\,\mathrm{A}\\
I_{\mathrm{D,\ min}} &= 5\,\mathrm{mA}\\
I_{\mathrm{L}} &= 1\,\mathrm{A} - 5\,\mathrm{mA} = 0{,}995\,\mathrm{A}\\
R_{\mathrm{L}} &= \frac{8\,\mathrm{V}}{0{,}995\,\mathrm{A}} \approx 8\,\Omega \Rightarrow R_{\mathrm{L,\ min}} \approx 8\,\Omega
\end{aligned}
$$

Siehe: Abschnitt 2.1.3

B.8 Einsatz einer Diode zur Leistungsreduzierung

a) Bei einem idealen Ventil: Nach dem Erreichen der Flussspannung verhält sich die Diode wie ein perfekter Leiter ohne Widerstand.
Da nur eine Halbwelle der Wechselspannung durchgelassen wird, halbiert sich die Leistung im Lastwiderstand. Die Diode selbst setzt keine Leistung um:

$$P_{\mathrm{D}} = 0\,\mathrm{W}$$

Die mittlere Leistung im Lastwiderstand beträgt:

$$P_{\mathrm{RL}} = \frac{1}{2} \cdot \frac{(U_{\mathrm{eff}})^2}{R_{\mathrm{L}}} = \frac{1}{2} \cdot \frac{(230\,\mathrm{V})^2}{1\,\mathrm{k\Omega}} = 26{,}45\,\mathrm{W}$$

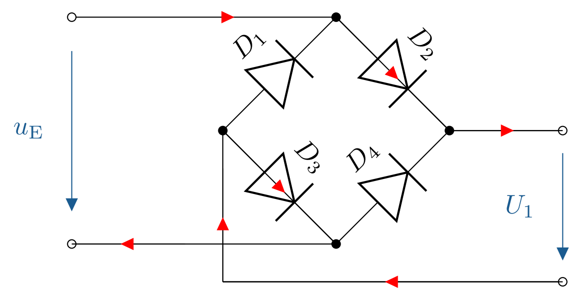
b) Bei überbrückter Diode liegt die volle Wechselspannung am Lastwiderstand an:

$$P_{\mathrm{RL}} = \frac{U_{\mathrm{eff}}^2}{R_{\mathrm{L}}} = \frac{230\,\mathrm{V}^2}{1\,\mathrm{k\Omega}} = 52{,}9\,\mathrm{W}$$

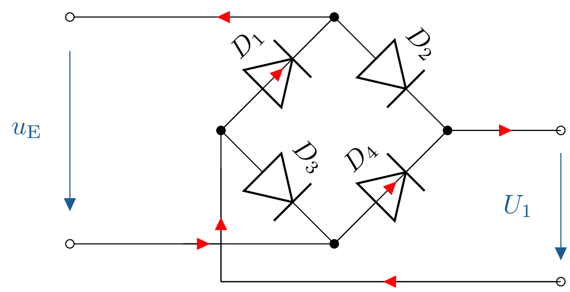
Siehe: Abschnitt 2.1.4

B.9 Spannungsverlauf in einer Brückengleichrichterschaltung

- Bei einer negativen Halbwelle fließt der Strom über D_2 und D_3.



- Bei einer positiven Halbwelle fließt der Strom über D_4 und D_1.



Siehe: Abschnitt 2.1.4

B.10 Berechnung des Vorwiderstands einer LED

Berechnung der Spannung am Vorwiderstand:

$$U_V = U_0 - U_D = 5\,\mathrm{V} - 2{,}2\,\mathrm{V} = 2{,}8\,\mathrm{V}$$

Berechnung des optimalen Vorwiderstands für $I_D = 30\,\mathrm{mA}$:

$$R_V = \frac{U_V}{I_D} = \frac{2{,}8\,\mathrm{V}}{30\,\mathrm{mA}} = 93{,}3\,\Omega$$

Berechnung des Stroms bei Auswahl eines konkreten Widerstandswertes:

$$I_D = \frac{U_V}{R} = \frac{2{,}8\,\mathrm{V}}{150\,\Omega} = 18{,}7\,\mathrm{mA}$$

Berechnung bei zwei Widerständen in Parallelschaltung mit zusätzlichem Serienwiderstand:

$$R_{ges} = \frac{150\,\Omega \cdot 180\,\Omega}{150\,\Omega + 180\,\Omega} + 15\,\Omega = 96{,}8\,\Omega$$
$$I_D = \frac{2{,}8\,\mathrm{V}}{96{,}8\,\Omega} = 28{,}9\,\mathrm{mA}$$

Alternative Reihenschaltung:

$$R_{ges} = 4 \cdot 15\,\Omega + 33\,\Omega = 93\,\Omega$$

Fazit: Ein kombinierter Widerstand aus $4 \times 15\,\Omega + 33\,\Omega = 93\,\Omega$ oder eine Parallelschaltung mit $150\,\Omega \,\|\, 180\,\Omega + 15\,\Omega = 96{,}8\,\Omega$ führt zu einem Strom nahe dem optimalen Wert von $30\,\mathrm{mA}$.
Empfohlene Wahl: Kombination mit Gesamtwiderstand nahe $93\,\Omega$.

Siehe: Abschnitt 2.1.3

B.11 Identifikation eines Bauteils anhand des Schaltzeichens

Beim dargestellten Bauelement handelt es sich um einen npn-Transistor. Siehe: Abschnitt 2.10

B.12 Ermittlung von Basis-Emitter-Spannung und Basisstrom

a) Diese Spannung ist eine materialbedingte Eigenschaft des Silizium-Transistors und bleibt nahezu konstant, unabhängig von der Basisspannung U_{B}. Ein npn-Transistor hat im leitenden Zustand eine typische Basis-Emitter-Spannung von:

$$U_{\mathrm{BE}} \approx 0{,}6\,\mathrm{V} \text{ bis } 0{,}7\,\mathrm{V}$$

Obwohl $U_0 = 5\,\mathrm{V}$ ist, begrenzt der Transistor die Spannung U_{BE} auf ca. 0,6 V, sobald der Transistor leitend ist. Der überschüssige Spannungsabfall wird durch andere Ströme und Bauteile der Schaltung ausgeglichen.

b)

$$I_{\mathrm{B}} = \frac{5\,\mathrm{V} - 0{,}6\,\mathrm{V}}{100\,\mathrm{k\Omega}} = 44\,\mu\mathrm{A}$$

Siehe: Abschnitt 2.2 und 2.14

B.13 Berechnung der Widerstände einer Transistorschaltung

a)

$$\frac{5\,\mathrm{V}}{2\,\mathrm{k\Omega} + 0{,}3\,\mathrm{k\Omega}} = \frac{U_1}{300\,\Omega} \Rightarrow U_1 = 300\,\Omega \cdot \frac{5\,\mathrm{V}}{2\,\mathrm{k\Omega} + 0{,}3\,\mathrm{k\Omega}} = 0{,}65\,\mathrm{V}$$

$$5\,\mathrm{V} = 0{,}65\,\mathrm{V} + U_{\mathrm{RV}} + 0{,}6\,\mathrm{V}$$

$$U_{\mathrm{RV}} = 5\,\mathrm{V} - 0{,}65\,\mathrm{V} - 0{,}6\,\mathrm{V} = 3{,}75\,\mathrm{V}$$

$$R_{\mathrm{V}} = \frac{3{,}75\,\mathrm{V}}{50\,\mu\mathrm{A}} = 75\,\mathrm{k\Omega}$$

b)

$$I_{\mathrm{C}} = \frac{15\,\mathrm{V} - 7{,}5\,\mathrm{V}}{500\,\Omega} = 15\,\mathrm{mA}$$

$$I_{\mathrm{E}} = I_{\mathrm{C}} + I_{\mathrm{B}} = 15\,\mathrm{mA} + 50\,\mu\mathrm{A} = 15{,}05\,\mathrm{mA}$$

$$\frac{I_{\mathrm{C}}}{I_{\mathrm{B}}} = \beta = 300$$

Siehe: Abschnitt 2.2 und 2.14

B.14 Schaltverhalten eines Transistors

a) Berechnung des Vorwiderstands für den Basisstrom:

$$R_{\mathrm{V}} = \frac{U}{I} = \frac{5\,\mathrm{V} - 0{,}6\,\mathrm{V}}{50\,\mu\mathrm{A}} = \frac{4{,}4\,\mathrm{V}}{50\,\mu\mathrm{A}} = 88\,\mathrm{k\Omega}$$

b) Berechnung des Kollektorstroms mit Stromverstärkungsfaktor:

$$\beta = \frac{I_\mathrm{C}}{I_\mathrm{B}}$$
$$I_\mathrm{C} = \beta \cdot I_\mathrm{B} = 180 \cdot 50\,\mu\mathrm{A} = 9\,\mathrm{mA}$$

Berechnung des Spannungsabfalls am Kollektorwiderstand:

$$U_\mathrm{RC} = I_\mathrm{C} \cdot R_\mathrm{C} = 9\,\mathrm{mA} \cdot 200\,\Omega = 1{,}8\,\mathrm{V}$$

Berechnung der Spannung zwischen Kollektor und Emitter:

$$U_\mathrm{CE} = 15\,\mathrm{V} - 1{,}8\,\mathrm{V} = 13{,}2\,\mathrm{V}$$

Siehe: Abschnitt 2.2 und 2.14

B.15 Transistorschaltung mit LED

Berechnung des Kollektorstroms bei bekannter Spannung und Lastwiderstand:

$$I_\mathrm{C} = \frac{U}{R}$$
$$I_\mathrm{C} = \frac{9\,\mathrm{V} - 0{,}3\,\mathrm{V}}{270\,\Omega} = 32{,}2\,\mathrm{mA}$$

Berechnung des benötigten Basisstroms mit dem Stromverstärkungsfaktor $\beta = 50$:

$$I_\mathrm{B} = \frac{I_\mathrm{C}}{\beta}$$
$$I_\mathrm{B} = \frac{32{,}2\,\mathrm{mA}}{50} = 0{,}644\,\mathrm{mA}$$

Spannungsabfall am Vorwiderstand:

$$U_\mathrm{V} = 0{,}644\,\mathrm{mA} \cdot 3{,}3\,\mathrm{k\Omega} = 2{,}12\,\mathrm{V}$$

Gesamte Basisspannung (inkl. U_BE):

$$U = U_\mathrm{V} + U_\mathrm{BE} = 2{,}12\,\mathrm{V} + 0{,}6\,\mathrm{V} = 2{,}72\,\mathrm{V}$$

Antwort: Die Basisspannung muss $\boxed{2{,}72\,\mathrm{V}}$ betragen, damit der Transistor durchschaltet.

Siehe: Abschnitt 2.2 und 2.14

B.16 Analyse des Arbeitsbereichs eines Transistors

a) Berechnung des Basisstroms über den Stromverstärkungsfaktor:

$$I_\mathrm{B} = \frac{I_\mathrm{C}}{\beta} = \frac{2\,\mathrm{mA}}{150} = 13{,}3\,\mu\mathrm{A}$$
$$I_\mathrm{E} = I_\mathrm{C} + I_\mathrm{B} = 2\,\mathrm{mA} + 13{,}3\,\mu\mathrm{A} = 2{,}013\,\mathrm{mA}$$
$$I_{\mathrm{R}_3} = 0{,}75 \cdot I_\mathrm{E} = 0{,}75 \cdot 2{,}013\,\mathrm{mA} = 1{,}51\,\mathrm{mA}$$
$$R_3 = \frac{3\,\mathrm{V}}{1{,}51\,\mathrm{mA}} = 1{,}99\,\mathrm{k\Omega}$$

b) Berechnung des Stroms durch R_2:

$$I_{R_2} = 9 \cdot I_\mathrm{B} = 9 \cdot 13{,}3\,\mu\mathrm{A} = 0{,}12\,\mathrm{mA}$$
$$R_2 = \frac{U_\mathrm{a} + U_\mathrm{BE}}{I_{\mathrm{R}_2}} = \frac{3\,\mathrm{V} + 0{,}6\,\mathrm{V}}{0{,}12\,\mathrm{mA}} = 30\,\mathrm{k\Omega}$$

Berechnung des Stroms durch R_1, welcher gleich I_{R_2} ist:

$$R_1 = \frac{U_\mathrm{BE}}{I_{\mathrm{R}_1}} = \frac{0{,}6\,\mathrm{V}}{0{,}12\,\mathrm{mA}} = 5{,}6\,\mathrm{k\Omega}$$

Siehe: Abschnitt 2.2.1

B.17 Berechnungen in einer Transistorschaltungen

a) Berechnung des maximalen Kollektorstroms bei idealem Transistor ($R_\mathrm{CE} = 0$):

$$I_\mathrm{C} = \frac{U_\mathrm{V}}{R_\mathrm{C}} = \frac{3\,\mathrm{V}}{7{,}5\,\Omega} = 400\,\mathrm{mA}$$

b) Bestimmung der Spannungen und des Kollektorstroms laut Kennlinienfeld:

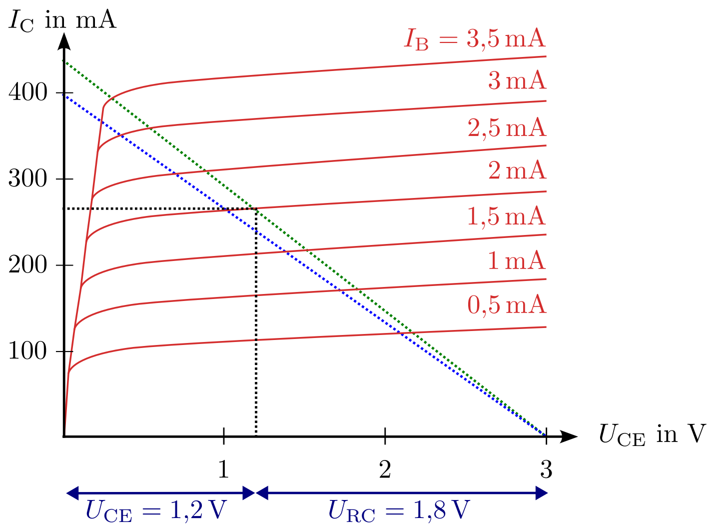


Figure B.2: Kennlinienfeld mit eingezeichneter Widerstandsgerade

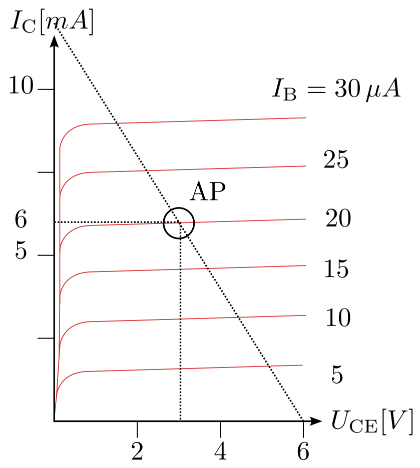
$$U_\mathrm{CE} = 1{,}2\,\mathrm{V}$$
$$U_\mathrm{RC} = 1{,}8\,\mathrm{V}$$
$$I_\mathrm{C} = \frac{U_\mathrm{RC}}{R_\mathrm{C}} = \frac{1{,}8\,\mathrm{V}}{7{,}5\,\Omega} = 240\,\mathrm{mA}$$

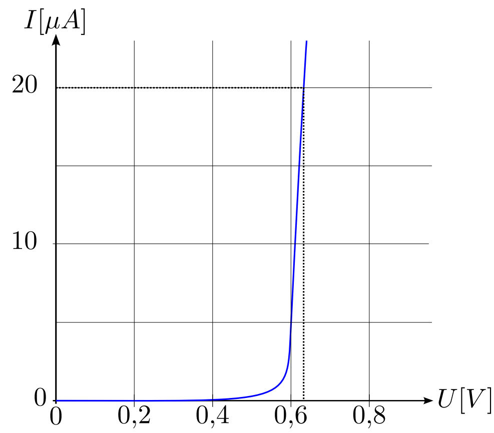
c) Berechnung des neuen Widerstandswerts bei gleichem I_C und neuer Spannung $U_\mathrm{RC} = 1{,}7\,\mathrm{V}$:

$$R_\mathrm{C} = \frac{U_\mathrm{RC}}{I_\mathrm{C}} = \frac{1{,}7\,\mathrm{V}}{240\,\mathrm{mA}} = 7{,}08\,\Omega$$

Siehe: Abschnitt 2.2.1

B.18 Berechnung der Stromverstärkung einer Transistorschaltung





Gegeben sind folgende Werte im Arbeitspunkt:

$$
\begin{aligned}
U_{CC} &= 6\,\mathrm{V}\\
U_{CE} &= 3\,\mathrm{V}\\
I_C &= 6\,\mathrm{mA}\\
U_{BE} &= 0{,}62\,\mathrm{V}\\
I_B &= 20\,\mu\mathrm{A}
\end{aligned}
$$

Berechnung der Stromverstärkung β:

$$\beta = \frac{I_C}{I_B} = \frac{6\,\mathrm{mA}}{20\,\mu\mathrm{A}} = 300$$

Siehe: Abschnitt 2.2.1

B.19 Verhalten eines Transistors bei Wechselspannung

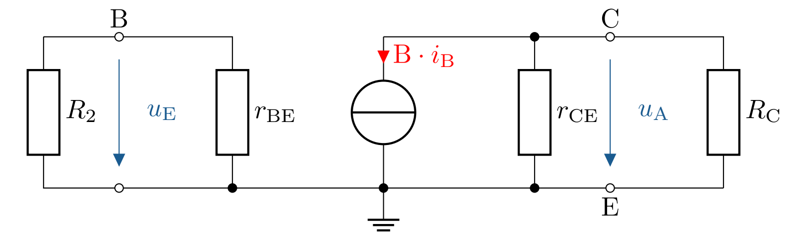
Wenn an einen Transistor eine reine Wechselspannung angelegt wird, tritt kein kontinuierlicher Verstärkungsbetrieb auf.

Das führt zu folgenden Effekten:

- Verzerrungen im Ausgangssignal, da nur die positiven Halbwellen verstärkt werden.
- Die negative Halbwelle wird abgeschnitten, was einer Gleichrichtung des Signals ähnelt.
- Es erfolgt keine lineare Verstärkung, da der Transistor zwischen Sperr- und Sättigungszustand wechselt.

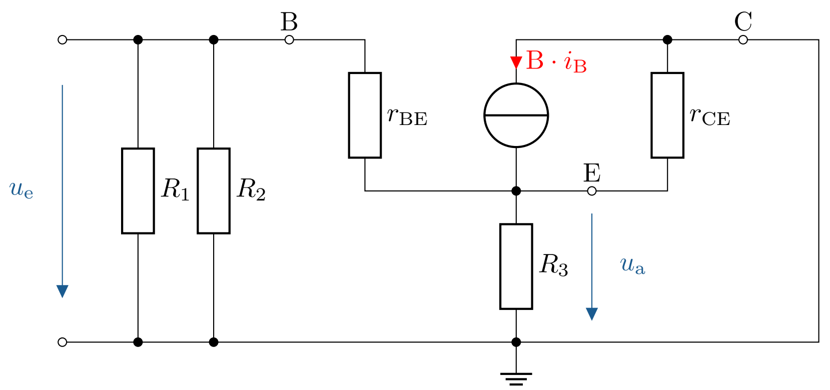
Siehe: Abschnitt 2.1.4

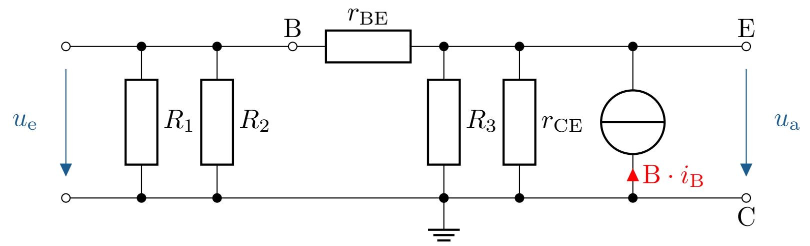
B.20 Erstellung eines Kleinsignal-Ersatzschaltbilds



Siehe: Abschnitt 2.18 und 2.24.

B.21 Erstellung eines Kleinsignal-Ersatzschaltbilds





Siehe: Abschnitt 2.18 und 2.24.

B.22 Bestimmung der Gate-Source-Spannung eines Feldeffekttransistors

In der Regel beträgt $U_{BE} = 0{,}7\,\mathrm{V}$.

Siehe: Abschnitt 2.3.1

B.23 Untersuchung des Stromflusses im Feldeffekttransistor (FET)

Zwischen Gate und Source.

Siehe: Abschnitt 2.3.1

B.24 Interpretation der Kennlinie eines Sperrschichttransistors

In Sättigungs- und Widerstandsbereich.

Siehe: Abschnitt 2.3.1

B.25 Erkennung von Dotierungsmustern bei Transistoren

a) n-Kanal selbstsperrend (Verarmungstyp) Siehe: Abschnitt 2.28

b) p-Kanal selbstsperrend (Verarmungstyp) Siehe: Abschnitt 2.28

c) n-Kanal selbstleitend (Anreicherungstyp) Siehe: Abschnitt 2.29

d) p-Kanal selbstleitend (Anreicherungstyp) Siehe: Abschnitt 2.29

B.26 Erklärung des Funktionsprinzips der CMOS-Technologie

e) In einer Schaltung werden sowohl p- als auch n-Kanal Transistoren verwendet

Siehe: Abschnitt 2.3.2

B.27 Transistor-Verbundverstärker

a)

$$I_{\mathrm{C}} = \frac{6\,\mathrm{V}}{10\,\mathrm{k\Omega}} = 0{,}6\,\mathrm{mA}$$

b)

$$I_{\mathrm{C,T2}} = 0{,}6\,\mathrm{mA}$$

c)

$$U_{\mathrm{R3}} = 0{,}6\,\mathrm{mA} \cdot 2\,\mathrm{k\Omega} = 1{,}2\,\mathrm{V}$$

d)

$$1{,}2\,\mathrm{V} + 0{,}6\,\mathrm{V} = 1{,}8\,\mathrm{V}$$

e)

$$I_{\mathrm{C,T1}} = \frac{8{,}2\,\mathrm{V}}{10\,\mathrm{k\Omega}} = 0{,}82\,\mathrm{mA}$$

f) Eine Erhöhung von I_{C} führt zu einem Spannungsanstieg über R_4, dadurch sinkt die Basisspannung von T_1, was I_{C} wieder reduziert. Die negative Rückkopplung stabilisiert den Arbeitspunkt.

g)

$$S = \frac{I_{\mathrm{C,T1}}}{U_{\mathrm{T}}} = \frac{0{,}82\,\mathrm{mA}}{26\,\mathrm{mA}} = 31{,}54\,\frac{\mathrm{mA}}{\mathrm{V}}$$

h)

$$A_{\mathrm{V}} = S \cdot (-R_{\mathrm{C}}) = -315$$

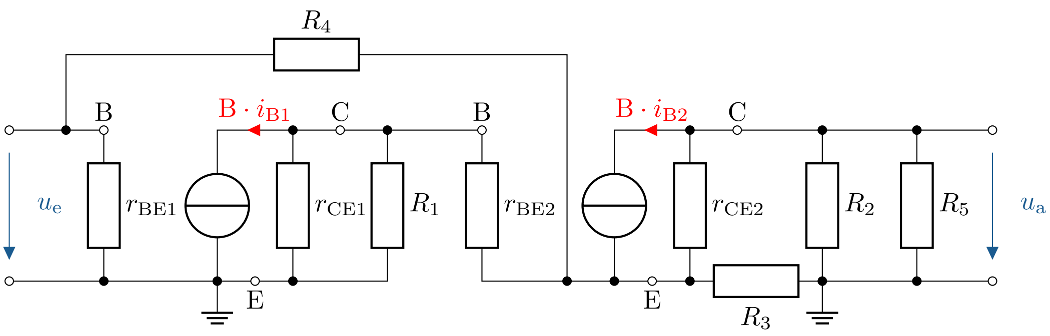
i)

$$A_{\mathrm{V}} = -\frac{R_{\mathrm{C}}}{R_{\mathrm{E}}} = \frac{10\,\mathrm{k\Omega}}{2\,\mathrm{k\Omega}} = -5$$

j)

$$-5 \cdot -315 = 1575 \approx 1600$$

k) Ersatzschaltbild



Siehe: Abschnitt 2.2.4

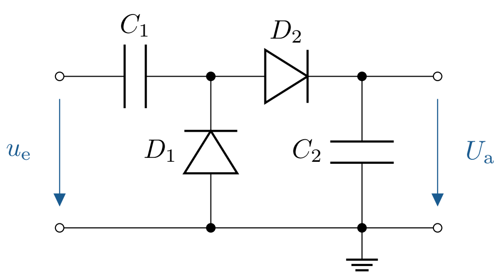
B.28 Gleichrichterschaltung

a)
- $U_{\mathrm{R1}} = 4{,}3\,\mathrm{V} \Rightarrow$ da 0,7 V über D_1 abfallen
- $U_{\mathrm{R2}} = -4{,}3\,\mathrm{V} \Rightarrow$ da gleiches für die negative Halbwelle
- $U_{\mathrm{R3}} = 8{,}6\,\mathrm{V} \Rightarrow$ da R_3 beide Spannungswerte aufnimmt

b) Die Kondensatoren C_1 und C_2 haben einerseits die Funktionen die Signale zu glätten, sowohl von der negativen als auch der positiven Halbwelle.

c) Der Vorteil dieser Gleichrichtung ist, das sowohl positive als auch negative Spannungen abgegriffen werden können.

d) Schaltung



Siehe: Abschnitt 2.7

B.29 Untersuchung eines einfachen Gleichrichters

a) Berechnung der Ausgangsspannung U_A

Die gegebene Schaltung besteht aus einer Diode in Vorwärtsrichtung und einem Lastwiderstand. Die Ausgangsspannung U_A entspricht der Eingangsspannung abzüglich der Diodendurchlasss-pannung:

$$U_\mathrm{A} = U_\mathrm{E} - U_\mathrm{f}$$

Einsetzen der Werte:

$$U_\mathrm{A} = 5\,\mathrm{V} - 0{,}7\,\mathrm{V} = 4{,}3\,\mathrm{V}$$

b) Bestimmung des Laststroms I_L

Der durch den Lastwiderstand R_L fließende Strom berechnet sich nach dem Ohmschen Gesetz:

$$I_\mathrm{L} = \frac{U_\mathrm{A}}{R_\mathrm{L}}$$

Einsetzen der Werte:

$$I_\mathrm{L} = \frac{4{,}3\,\mathrm{V}}{1\,\mathrm{k\Omega}} = 4{,}3\,\mathrm{mA}$$

Siehe: Abschnitt 2.1.4

B.30 Analyse einer Spannungsverdoppler-Schaltung

a) Funktionsprinzip der Delon-Schaltung
Die Schaltung arbeitet als Kaskaden-Gleichrichter. Während der positiven Halbwelle der Wechselspannung lädt sich C_1 über D_1 auf die Eingangsspitzen-Spannung $\hat{U}_\mathrm{E}$ auf. Während der negativen Halbwelle lädt sich C_2 über D_2 auf. Dadurch addieren sich die Spannungen von C_1 und C_2, wodurch am Ausgang eine theoretische Gleichspannung von ca. $2 \cdot \hat{U}_\mathrm{E}$ entsteht.

b) Berechnung der Leerlauf-Ausgangsspannung U_A Die Eingangsspannung ist als Effektivwert U_E gegeben. Die Spitzenwertspannung beträgt:

$$\hat{U}_\mathrm{E} = \sqrt{2} \cdot U_\mathrm{E}$$

Da die Schaltung eine Spannungsverdopplung erzeugt, ergibt sich die Leerlauf-Ausgangsspannung:

$$U_\mathrm{A} = 2 \cdot (\hat{U}_\mathrm{E} - U_\mathrm{D}) = 2 \cdot (\sqrt{2} \cdot U_E - 0{,}7\,\mathrm{V})$$

Für $U_\mathrm{E} = 110\,\mathrm{V}$:

$$U_\mathrm{A} = 2 \cdot (\sqrt{2} \cdot 110\,\mathrm{V} - 0{,}7\,\mathrm{V}) \approx 308\,\mathrm{V}$$

Für $U_\mathrm{E} = 230\,V$:

$$U_\mathrm{A} = 2 \cdot (\sqrt{2} \cdot 230\,\mathrm{V} - 0{,}7\,\mathrm{V}) \approx 644\,\mathrm{V}$$

c) Verhalten unter Lastbedingungen Im Leerlauf bleibt die Ausgangsspannung hoch. Unter Last fällt die Spannung ab, abhängig vom Innenwiderstand der Kondensatoren, der Belastung R_Last und den Diodenverlusten. Der Spannungsabfall hängt von der Restwelligkeit ab, die durch die Lade- und Entladezeiten der Kondensatoren beeinflusst wird. Ein großer Laststrom reduziert die Spannung, da die Kondensatoren sich schneller entladen.

Faktoren, die die Spannung beeinflussen:

- Innenwiderstand der Dioden
- Kapazität der Kondensatoren C
- Belastung durch R_Last
- Netzfrequenz (50 Hz oder 60 Hz)

B.31 Grundverhalten eines npn-Bipolartransistors

a) Berechnung des Kollektorstroms I_C Der Kollektorstrom eines Bipolartransistors ergibt sich aus der Beziehung:

$$I_C = \beta \cdot I_B$$

mit den gegebenen Werten:

$$I_C = 100 \cdot 20\,\mu\text{A} = 2\,\text{mA}$$

b) Bestimmung der Kollektor-Emitter-Spannung U_{CE} Die Kollektor-Emitter-Spannung ergibt sich aus der Gleichung:

$$U_{CE} = U_{CC} - I_C \cdot R_4$$

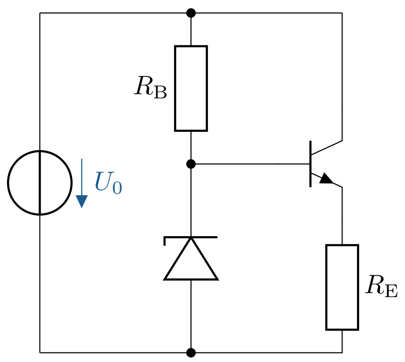
Einsetzen der Werte:

$$U_{CE} = 12\,\text{V} - (2\,\text{mA} \cdot 1\,\text{k}\Omega) = 12\,\text{V} - 2\,\text{V} = 10\,\text{V}$$

Siehe: Abschnitt 2.10

B.32 Zeichnung und Berechnung einer Kollektorschaltung

a) Schaltung:



b) Berechnung der Emitterspannung U_E
Die Zenerdiode fixiert die Basisspannung des Transistors auf $U_Z = 10\,\text{V}$. Die Basis-Emitter-Spannung beträgt $U_{BE} = 0{,}7\,\text{V}$.

$$U_E = U_Z - U_{BE} = 10\,\text{V} - 0{,}7\,\text{V} = 9{,}3\,\text{V}$$
$$I_E = \frac{U_E}{R_E} = \frac{9{,}3\,\text{V}}{1\,\text{k}\Omega} = 9{,}3\,\text{mA}$$

c) Berechnung der Leistung am Emitterwiderstand P_{RE}

$$P_{RE} = I_E^2 \cdot R_E = (9{,}3\,\text{mA})^2 \cdot 1\,\text{k}\Omega = 86{,}49\,\text{mW}$$

Siehe: Abschnitt 2.23